

Java™ 2 Enterprise Edition Management Specification

JSR-77

Public Draft

Hans Hrasna
Sun Microsystems, Inc.

Java(TM) 2 Platform, Enterprise Edition, Management Specification ("Specification")**Version: 1.0****Status: Pre-FCS****Release: December 17, 2001**

Copyright 2001 Sun Microsystems, Inc.
901 San Antonio Road, Palo Alto, California 94303, U.S.A.
All rights reserved.

NOTICE

The Specification is protected by copyright and the information described therein may be protected by one or more U.S. patents, foreign patents, or pending applications. Except as provided under the following license, no part of the Specification may be reproduced in any form by any means without the prior written authorization of Sun Microsystems, Inc. ("Sun") and its licensors, if any. Any use of the Specification and the information described therein will be governed by the terms and conditions of this license and the Export Control and General Terms as set forth in Sun's website Legal Terms. By viewing, downloading or otherwise copying the Specification, you agree that you have read, understood, and will comply with all of the terms and conditions set forth herein.

Subject to the terms and conditions of this license, Sun hereby grants you a fully-paid, non-exclusive, non-transferable, worldwide, limited license (without the right to sublicense) under Sun's intellectual property rights to review the Specification internally for the purposes of evaluation only. Other than this limited license, you acquire no right, title or interest in or to the Specification or any other Sun intellectual property. The Specification contains the proprietary and confidential information of Sun and may only be used in accordance with the license terms set forth herein. This license will expire ninety (90) days from the date of Release listed above and will terminate immediately without notice from Sun if you fail to comply with any provision of this license. Upon termination, you must cease use of or destroy the Specification.

TRADEMARKS

No right, title, or interest in or to any trademarks, service marks, or trade names of Sun or Sun's licensors is granted hereunder. Sun, Sun Microsystems, the Sun logo, Java, J2EE, and the Java Coffee Cup Logo are trademarks or registered trademarks of Sun Microsystems, Inc. in the U.S. and other countries.

DISCLAIMER OF WARRANTIES

THE SPECIFICATION IS PROVIDED "AS IS" AND IS EXPERIMENTAL AND MAY CONTAIN DEFECTS OR DEFICIENCIES WHICH CANNOT OR WILL NOT BE CORRECTED BY SUN. SUN MAKES NO REPRESENTATIONS OR WARRANTIES, EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO, WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, OR NON-INFRINGEMENT THAT THE CONTENTS OF THE SPECIFICATION ARE SUITABLE FOR ANY PURPOSE OR THAT ANY PRACTICE OR IMPLEMENTATION OF SUCH CONTENTS WILL NOT INFRINGE ANY THIRD PARTY PATENTS, COPYRIGHTS, TRADE SECRETS OR OTHER RIGHTS. This document does not represent any commitment to release or implement any portion of the Specification in any product.

THE SPECIFICATION COULD INCLUDE TECHNICAL INACCURACIES OR TYPOGRAPHICAL ERRORS. CHANGES ARE PERIODICALLY ADDED TO THE INFORMATION THEREIN; THESE CHANGES WILL BE INCORPORATED INTO NEW VERSIONS OF THE SPECIFICATION, IF ANY. SUN MAY MAKE IMPROVEMENTS AND/OR CHANGES TO THE PRODUCT(S) AND/OR THE PROGRAM(S) DESCRIBED IN THE SPECIFICATION AT ANY TIME. Any use of such changes in the Specification will be governed by the then-current license for the applicable version of the Specification.

LIMITATION OF LIABILITY

TO THE EXTENT NOT PROHIBITED BY LAW, IN NO EVENT WILL SUN OR ITS LICENSORS BE LIABLE FOR ANY DAMAGES, INCLUDING WITHOUT LIMITATION, LOST REVENUE, PROFITS OR DATA, OR FOR SPECIAL, INDIRECT, CONSEQUENTIAL, INCIDENTAL OR PUNITIVE DAMAGES, HOWEVER CAUSED AND REGARDLESS OF THE THEORY OF LIABILITY, ARISING OUT OF OR RELATED TO ANY FURNISHING, PRACTICING, MODIFYING OR ANY USE OF THE SPECIFICATION, EVEN IF SUN AND/OR ITS LICENSORS HAVE BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES.

You will indemnify, hold harmless, and defend Sun and its licensors from any claims based on your use of the Specification for any purposes other than those of internal evaluation, and from any claims that later versions or releases of any Specification furnished to you are incompatible with the Specification provided to you under this license.

RESTRICTED RIGHTS LEGEND

If this Software is being acquired by or on behalf of the U.S. Government or by a U.S. Government prime contractor or subcontractor (at any tier), then the Government's rights in the Software and accompanying documentation shall be only as set forth in this license; this is in accordance with 48 C.F.R. 227.7201 through 227.7202-4 (for Department of Defense (DoD) acquisitions) and with 48 C.F.R. 2.101 and 12.212 (for non-DoD acquisitions).

REPORT

You may wish to report any ambiguities, inconsistencies or inaccuracies you may find in connection with your evaluation of the Specification ("Feedback"). To the extent that you provide Sun with any Feedback, you hereby: (i) agree that such Feedback is provided on a non-proprietary and non-confidential basis, and (ii) grant Sun a perpetual, non-exclusive, worldwide, fully paid-up, irrevocable license, with the right to sublicense through multiple levels of sublicensees, to incorporate, disclose, and use without limitation the Feedback for any purpose related to the Specification and future versions, implementations, and test suites thereof.

Contents

JSR77.1 Introduction	9
JSR77.1.1 Overview	9
JSR77.1.2 Scope	10
JSR77.1.3 Organization	10
JSR77.1.4 Acknowledgments	11
JSR77.2 J2EE Management Model	13
JSR77.2.1 Overview	13
JSR77.2.2 Elements of the J2EE Management Model	14
JSR77.2.3 Interpreting the UML Diagrams	14
JSR77.2.3.1 Supplier Cardinality	16
JSR77.2.3.2 J2EEApplication managed object example	16
JSR77.2.4 Extending the models	18
JSR77.3 Managed Objects	19
JSR77.3.1 J2EEManagedObject	20
JSR77.3.1.1 Attribute Detail	21
JSR77.3.2 J2EEDomain extends J2EEManagedObject	26
JSR77.3.2.1 Attribute Detail	27
JSR77.3.3 J2EEServer extends J2EEManagedObject	28
JSR77.3.3.1 Attribute Detail	29
JSR77.3.4 JVM extends J2EEManagedObject	30
JSR77.3.4.1 Attribute Detail	30
JSR77.3.5 Node extends J2EEManagedObject	31
JSR77.3.5.1 Attribute Detail	31
JSR77.3.6 J2EEDeployedObject extends J2EEManagedObject	32
JSR77.3.7 J2EEApplication extends J2EEDeployedObject	32
JSR77.3.7.1 Attribute Detail	33
JSR77.3.8 J2EEModule extends J2EEDeployedObject	33
JSR77.3.8.1 Attribute Detail	34
JSR77.3.9 EJBModule extends J2EEModule	34
JSR77.3.9.1 Attribute Detail	34
JSR77.3.10EJB extends J2EEManagedObject	35

JSR77.3.11MessageDrivenBean extends EJB	35
JSR77.3.12EntityBean extends EJB	35
JSR77.3.13SessionBean extends EJB	35
JSR77.3.14StatefulSessionBean extends SessionBean	36
JSR77.3.15StatelessSessionBean extends SessionBean	36
JSR77.3.16WebModule extends J2EEModule	36
JSR77.3.16.1 Attribute Detail	36
JSR77.3.17Servlet extends J2EEManagedObject	36
JSR77.3.18ConnectorModule extends J2EEModule	36
JSR77.3.18.1 Attribute Detail	37
JSR77.3.19ResourceAdapter extends J2EEManagedObject	37
JSR77.3.20J2EEResource extends J2EEManagedObject	37
JSR77.3.21JavaMail extends J2EEResource	38
JSR77.3.22JMS extends J2EEResource	38
JSR77.3.23JNDI extends J2EEResource	38
JSR77.3.24JTA extends J2EEResource	38
JSR77.3.25RMI_IOP extends J2EEResource	38
JSR77.3.26URL extends J2EEResource	38
JSR77.3.27JDBC extends J2EEResource	39
JSR77.3.27.1 Attribute Detail	39
JSR77.3.28JDBCDataSource extends J2EEManagedObject	40
JSR77.3.28.1 Attribute Detail	40
JSR77.3.29JDBCDriver extends J2EEManagedObject	40
JSR77.4 Events	41
JSR77.4.1 EventProvider	41
JSR77.4.1.1 Attribute Detail	41
JSR77.4.2 Notification	42
JSR77.4.2.1 Attribute Detail	42
JSR77.4.3 Event Notification	44
JSR77.4.3.1 javax.management.Notification	44
JSR77.5 State Management	45
JSR77.5.1 StateManageable	45
JSR77.5.1.1 Attribute Detail	46
JSR77.5.1.2 Operation Detail	47
JSR77.6 Performance Monitoring	49

JSR77.6.1 Performance Data Framework	49
JSR77.6.2 StatisticsProvider	49
JSR77.6.2.1 Attribute Detail	50
JSR77.6.3 package javax.management.j2ee.statistics	51
JSR77.6.4 Statistic	51
JSR77.6.4.1 Operation Detail	52
JSR77.6.5 CountStatistic extends Statistic	52
JSR77.6.5.1 Operation Detail	53
JSR77.6.6 TimeStatistic extends Statistic	53
JSR77.6.6.1 Operation Detail	53
JSR77.6.7 RangeStatistic extends Statistic	54
JSR77.6.7.1 Operation Detail	54
JSR77.6.8 BoundaryStatistic extends Statistic	55
JSR77.6.8.1 Operation Detail	55
JSR77.6.9 BoundedRangeStatistic extends RangeStatistic, BoundaryStatistic	56
JSR77.6.9.1 Operation Detail	56
JSR77.6.10Stats	57
JSR77.6.10.1 Operation Detail	57
JSR77.6.11EJBStats extends Stats	58
JSR77.6.11.1 Operation Detail	58
JSR77.6.12EntityBeanStats extends EJBStats	59
JSR77.6.12.1 Operation Detail	59
JSR77.6.13MessageDrivenBeanStats extends EJBStats	59
JSR77.6.13.1 Operation Detail	59
JSR77.6.14SessionBeanStats extends EJBStats	60
JSR77.6.14.1 Operation Detail	60
JSR77.6.15StatefulSessionBeanStats extends SessionBeanStats	60
JSR77.6.15.1 Operation Detail	60
JSR77.6.16StatelessSessionBeanStats extends SessionBeanStats	60
JSR77.6.17JavaMailStats extends Stats	61
JSR77.6.17.1 Operation Detail	61
JSR77.6.18JDBCStats extends Stats	62
JSR77.6.18.1 Operation Detail	62
JSR77.6.19JDBCConnectionStats extends Stats	63
JSR77.6.19.1 Operation Detail	63
JSR77.6.20JDBCConnectionPoolStats extends JDBCConnectionStats	64

JSR77.6.20.1 Operation Detail	64
JSR77.6.21JMSSStats extends Stats	65
JSR77.6.21.1 Operation Detail	65
JSR77.6.22JMSSConnectionStats extends Stats	66
JSR77.6.22.1 Operation Detail	66
JSR77.6.23JMSSessionStats extends Stats	66
JSR77.6.23.1 Operation Detail	66
JSR77.6.24JMSEndpointStats extends Stats	68
JSR77.6.24.1 Operation Detail	68
JSR77.6.25JMSProducerStats extends JMSEndpointStats	69
JSR77.6.25.1 Operation Detail	69
JSR77.6.26JMSSConsumerStats extends JMSEndpointStats	69
JSR77.6.26.1 Operation Detail	69
JSR77.6.27JTASStats extends Stats	70
JSR77.6.27.1 Operation Detail	70
JSR77.6.28JVMStats extends Stats	71
JSR77.6.28.1 Operation Detail	71
JSR77.6.29 ServletStats extends Stats	72
JSR77.6.29.1 Operation Detail	72
JSR77.6.30URLStats extends Stats	72
JSR77.6.30.1 Operation Detail	72
JSR77.7 J2EE Management EJB Component	73
JSR77.7.1 MEJB Implementation requirements	73
JSR77.7.1.1 JMX requirements	74
JSR77.7.1.2 Remote Interface	74
JSR77.7.1.3 Home Interface	74
JSR77.7.1.4 Naming Requirements	74
JSR77.7.1.5 Event Listener Requirements	75
JSR77.7.2 Using the Management EJB component	76
JSR77.7.2.1 Creating a MEJB Session Object	76
JSR77.7.2.2 Finding managed objects of a certain type	76
JSR77.7.2.3 Retrieving attribute information	77
JSR77.7.2.4 Invoking methods	78
JSR77.7.2.5 Registering an event notification listener	78
JSR77.7.3 package javax.management.j2ee	79
JSR77.7.3.1 Interface ListenerRegistration	79
JSR77.7.3.2 Interface Management	82

JSR77.7.3.3 Interface ManagementHome	90
JSR77.8 J2EE Management SNMP	91
JSR77.8.1 Abstract.	91
JSR77.8.2 Introduction.	91
JSR77.8.3 The SNMP Management Framework.....	92
JSR77.8.4 Overview.	93
JSR77.8.5 The Structure of the MIB	94
JSR77.8.6 Definitions	96
JSR77.8.7 IANA Considerations	128
JSR77.8.8 Security Considerations	128
JSR77.8.9 Intellectual Property	129
JSR77.8.10 Acknowledgments	129
JSR77.8.11 References.	130
JSR77.9 J2EE Management CIM.....	131
JSR77.9.1 Introduction.	131
JSR77.9.2 Managed Objects	131
JSR77.9.2.1 J2EEDomain	131
JSR77.9.2.2 J2EEServer	132
JSR77.9.3 Events	133
JSR77.9.3.2 Modeling J2EE Events in CIM	134
JSR77.9.3.3 Handling subscriptions	134
JSR77.9.4 State Management	135
JSR77.9.5 Appendix.	136
JSR77.9.5.1 J2EE Core Model.	136
JSR77.9.5.2 J2EE Events and Indications Model	146
JSR77.9.5.3 J2EE State Model	148
JSR77.9.5.4 J2EE Statistics Model	150

CHAPTER JSR77.1

Introduction

The increasing popularity of the Java™ 2 Enterprise Edition (J2EE™) platform has mandated the specification of a common framework for the delivery of Java enterprise management and monitoring services. The goal of JSR77 is to abstract the fundamental manageable aspects of the J2EE architecture to provide a well defined model for implementing instrumentation and information access. In addition, this specification defines a standardized API for interoperating with J2EE™ components that participate in the monitoring and control of the platform's resources.

JSR77.1.1 Overview

This specification defines a management information model for the J2EE platform, the J2EE Management Model. This specification also includes standard mappings of the model to the Common Information Model (CIM), an SNMP Management Information Base (MIB), and to a Java API as a server resident Enterprise JavaBeans™ (EJB™) component, the J2EE Management EJB (MEJB) component. The MEJB component provides interoperable remote access to the model from any standard J2EE application.

JSR77.1.2 Scope

The scope of this version of the specification is limited to the management of J2EE servers, application components that are deployed onto a J2EE server and resources which provide the J2EE Standard Services. The scope of the information and operations required by the specification is limited to that required to enable basic functionality for the following:

- Discovery - discovering and navigating the managed objects on a managed system
- Events - receiving notifications of important events that occur on managed objects
- State - observing and controlling the running state of a managed objects
- Performance - monitoring basic performance statistics of managed objects

The following management categories are also considered important but were deemed out of scope for this version of the specification:

- Security management
- Accounting
- Configuration management
- Fault Management/Recovery

JSR77.1.3 Organization

- Chapter 2 introduces the J2EE Management Model and provides definitions of the terms used to describe the model.
- Chapter 3 specifies the structure, format, and semantics of managed objects required by this specification.
- Chapter 4 specifies a standard extensible event model for the J2EE Management Model.
- Chapter 5 specifies a standard state management model for the J2EE Management Model.

- Chapter 6 specifies the performance data requirements of the J2EE Management Model.
- Chapter 7 specifies the J2EE Management EJB component (MEJB).
- Chapter 8 specifies the SNMP MIB mapping of the J2EE Management Model.
- Chapter 9 specifies the CIM mapping of the J2EE Management Model.

JSR77.1.4 Acknowledgments

This specification was developed under the Java Community Process 2.0 as JSR-077 and includes contributions from numerous groups at Sun and at partner companies. We would like to thank the members of the JSR-077 Expert Group in particular for their contributions:

- Altaworks Corporation - Ralph Beck
- BEA Systems - Vadim Draluk, Steve Fanshier
- Borland Software Corporation - Geoff Bullen, Michael Stern
- Flashline - Gary Baney
- Gemstone Systems, Inc. - Chris Jacobson
- IBM - Leigh Williamson
- IONA Technologies - Brian Joyce
- Informix Software - Tony Daniel
- JBoss - Marc Fluery
- Laminar Software - Anand Byrappagari
- Lutris Technologies - David Simons
- Pramati Technologies - Ramesh Loganathan
- Sitraka - Alan Armstrong, Doug Doe
- Sun Microsystems Inc. - Rebecca Searls

CHAPTER JSR77.2

J2EE Management Model

This chapter describes the J2EE Management Model design and how it is applied to managed systems.

JSR77.2.1 Overview

The J2EE Management Model is a specification of the attributes, operations and architecture of the managed objects required by compliant J2EE platform implementations. The model is designed to be interoperable with a variety of industry standard management systems and protocols. While the diagrams and textual descriptions that comprise the J2EE Management Model closely resemble Java classes, they are not specifications of Java class types or Java class inheritance hierarchies and do not represent requirements of the class names or class hierarchies of a particular implementation. They are specifications of the attributes, operations and associations required by instances of the specified managed object types.

Standard mappings of the model to the CIM/WBEM Common Information Model (CIM) , the SNMP Management Information Base (MIB) and a Java API are also provided by this specification. The Management EJB (MEJB) component provides an interoperable interface and remote transport protocol to the model from J2EE components.

Support of the MEJB component is required by all compliant implementations. Support for SNMP and CIM/WBEM implementations of the model is optional.

JSR77.2.2 Elements of the J2EE Management Model

The elements that make up the J2EE Management Model are specified by the following chapters:

- Chapter JSR77.3, “Managed Objects” specifies the managed object types required by the J2EE Management Model. All of the managed objects described in this chapter must be implemented by the J2EE platform provider.
- Chapter JSR77.4, “Events” specifies the EventProvider model required by managed objects that optionally support event notification.
- Chapter JSR77.5, “State Management” specifies the StateManageable model required by managed objects that optionally support state manipulation.
- Chapter JSR77.6, “Performance Monitoring” specifies the StatisticProvider model required by managed objects that optionally support performance data.

The specification of the models includes the textual specification as well as the UML diagrams of the metadata.

JSR77.2.3 Interpreting the UML Diagrams

The elements specified by the managed object UML diagrams consist of:

- A name which identifies the managed object types and supertypes.
 - The use of supertypes in the managed object model does not imply the capability of polymorphism. Inheritance in the models is used simply as a convenient way to express the common attributes and operations shared by a manageable type. The set of standard manageable types is limited to the types specified by the model at the end of the hierarchy of inheritance of a particular model. The standard manageable types are:
 - J2EEDomain
 - J2EEServer
 - J2EEApplication
 - EJBModule
 - WebModule
 - ConnectorModule
 - EntityBean
 - StatefulSessionBean
 - StatelessSessionBean

- MessageDrivenBean
 - Servlet
 - ResourceAdapter
 - JavaMail
 - JDBC
 - JDBCDataSource
 - JDBCDriver
 - JMS
 - JNDI
 - JTA
 - RMI_IIOP
 - URL
 - JVM
 - Node
- Attributes which contain data about the managed object.
 - For increased interoperability with various management systems, attribute types in the specified model are limited to String, boolean, int, long and double.
 - All specified attributes must implemented as readable and may also be implemented as writable.
- Attributes which identify managed objects.
 - The attribute which identifies a managed object and attributes which identify associated managed objects are specified with a specialized string type referred to in this specification as OBJECT_NAME. The type OBJECT_NAME refers to a formatted string whose syntax is defined by the specification of the name attribute of the J2EEManagedObject model (see section JSR77.3.1.1.1 on page 21). The OBJECT_NAME string type provides a unique identifier for a managed object within a J2EE management domain. Every managed object has a name attribute whose value complies with the OBJECT_NAME syntax.
- Operations which describe the functions required by a managed object type.
 - All operations specified for a managed object type must be invocable by a management client.

JSR77.2.3.1 Supplier Cardinality

In each of the attributes in the model that provide a list of managed objects there is also a specification of supplier cardinality. Supplier cardinality specifies the minimum and maximum number of items that can be expected to exist in a list. Supplier cardinality is specified as:

min..max

Where *min* can be any whole number and *max* can be any whole number or an asterisk '*', which indicates that there is no maximum limit.

JSR77.2.3.2 J2EEApplication managed object example

The following example of accessing a J2EEApplication managed object instance illustrates how the J2EE Management Model specification of the J2EEApplication type translates to an actual management system implementation which is being access through the J2EE Management EJB component.

The specification for the J2EEApplication managed object type (section JSR77.3.7 on page 32) indicates that J2EEApplication extends J2EEDeployedObject which extends J2EEManagedObject.

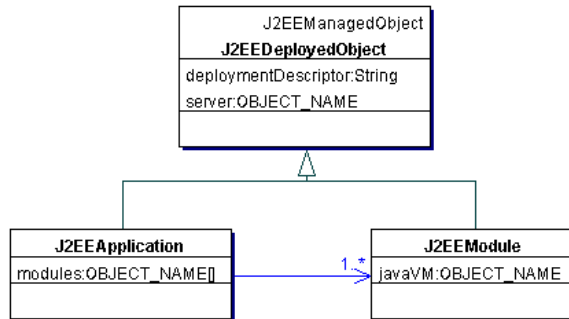


Figure JSR77.2.1 J2EEApplication model

This indicates that a managed object instance of the J2EEApplication type must support the following attributes:

- **OBJECT_NAME** name
- **boolean** stateManageable
- **boolean** statisticsProvider
- **boolean** eventProvider

- **String deploymentDescriptor**
- **OBJECT_NAME server**
- **OBJECT_NAME[] modules**

If any of the boolean attributes is true it indicates that this J2EEApplication managed object also includes the required attributes and operations of the respective models StateManageable, StatisticsProvider, or EventProvider.

The following code fragment shows how a J2EE component accesses a managed object of the J2EEApplication type through the J2EE Management EJB component:

Code Example 2.2

```
// Note: applicationName is the javax.management.ObjectName for the application
//
// get the value of an attribute which is a primitive type
// in this case, the deployment descriptor for the application
String deploymentDescriptor = (String) mejb.getAttribute(applicationName,
"deploymentDescriptor");

// get the modules for this application
ObjectName[] modules =
(ObjectName[])mejb.getAttribute(applicationName, "modules");
```

Now that the component has the array of ObjectName instances for all of the J2EEModule managed objects associated with this application, we can use them to access the module's attributes and operations the same way we accessed the application:

Code Example 2.3

```
// get the deployment descriptors for the modules
String[] moduleDeploymentDescriptors = new String[modules.length];
for (int i=0; i < modules.length; i++) {
    moduleDeploymentDescriptors[i] = (String) mejb.getAttribute(modules[i],
"deploymentDescriptor");
}
```

JSR77.2.4 Extending the models

It is permitted to extend the functionality of any of the models in the J2EE Management model by adding vendor specific attributes, operations and associations. The attributes, operations and associations specified by the models must always be supported exactly as specified.

CHAPTER JSR77.3

Managed Objects

The term *managed object* refers to the definition of a unit of management information. Management instrumentation provides the “glue” which takes the information available on an entity to be managed and makes it appear as a collection of managed objects.

This chapter contains the models and metamodels which specify the format, semantics and structural relationship of the managed objects required by all compliant implementations of this specification.

JSR77.3.1 J2EEManagedObject

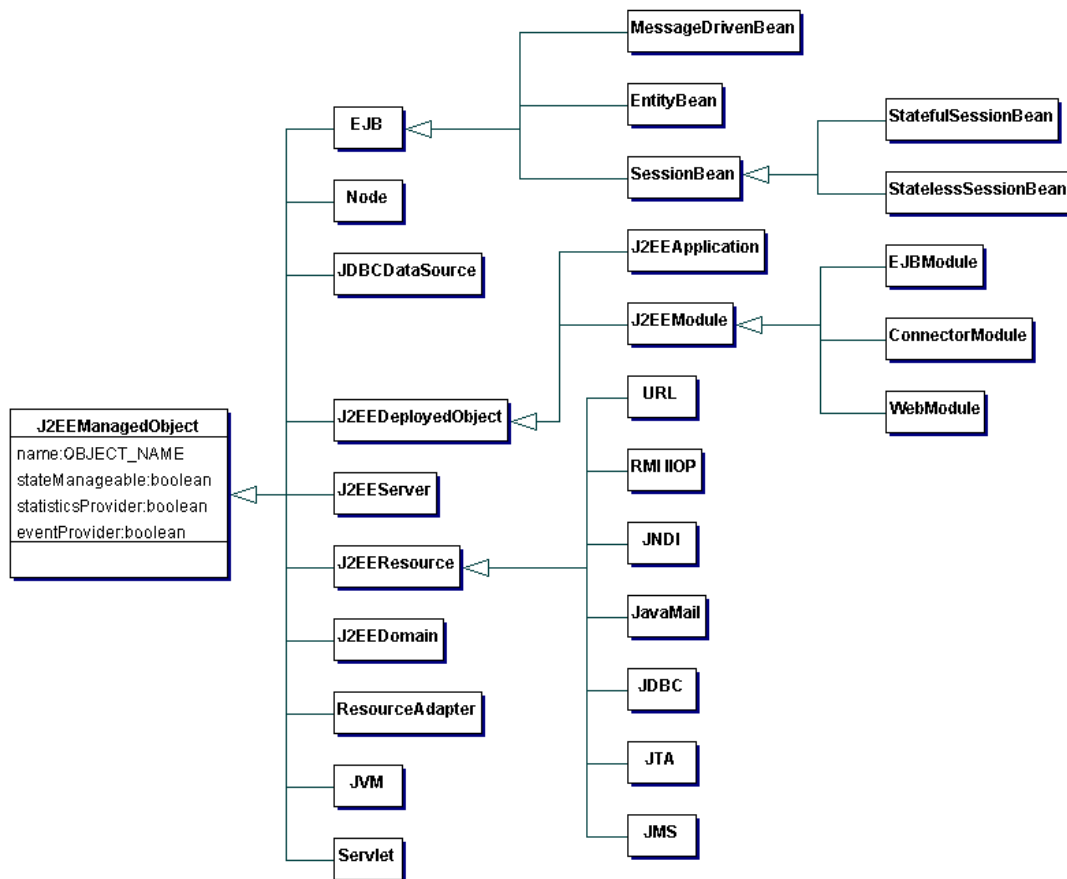


Figure JSR77.3.1 J2EEManagedObject detail with inheritance overview

The J2EEManagedObject model is the base model of all managed objects in the J2EE Management Model. All managed objects in the J2EE Platform must include the attributes of the J2EEManagedObject model. All managed objects must have a unique name.

Any managed object may optionally implement the metadata specified by the StateManageable model (see “StateManageable” on page 45), which enables the managed object to provide state management operations.

Any managed object may optionally implement the metadata specified by the StatisticsProvider model (see “StatisticsProvider” on page 49), which provides access to the performance statistics for the managed object.

Any managed object may optionally implement the EventProvider model, which indicates it emits events which implement the J2EEManagementEvent model (see “EventProvider” on page 41).

JSR77.3.1.1 Attribute Detail

JSR77.3.1.1.1 *name*

OBJECT_NAME *name*

The name of the managed object. The name attribute is of the type OBJECT_NAME which is a string that complies with the syntax specified for a J2EEManagedObject name in this section. The name string must not be null. The name must be unique within the management domain. Management applications use this name to identify managed objects, for example identifying the source of events.

The J2EEManagedObject name consists of two parts:

- A domain name
- An unordered set of key properties, which must include the type, and name key properties.

The J2EEManagedObject name has the following syntax:

[domainName];type=value,name=value[,property=value]*

JSR77.3.1.1.1.1 Domain Name

The domain name is a case-sensitive string. It provides a structure for the naming space within a J2EE management domain. How the domain name is structured is application-dependent. The domain name string may contain any characters, including spaces, except those which are object name separators or wildcards, namely the colon, comma, equals sign, asterisk or question mark (;,=*,?). The domain name is treated as a whole, therefore any semantic sub-definitions within the string are opaque to a J2EE Management implementation. The value of the name property for a J2EEDomain managed object must be equivalent to the domain name.

JSR77.3.1.1.1.2 Key Property List

The key property list enables you to assign unique names to the managed objects of a given domain. A key property is a property-value pair, where the property does not need to correspond to an actual attribute of a managed object. The key property list must contain at least the mandatory key properties. It may contain any number of optional key properties whose order is not significant. Keys and values are subject to the same character restrictions as domain names.

JSR77.3.1.1.1.3 Mandatory Key Properties

The following key properties are required in the key property list for every managed object name:

- type - specifies the type of the managed object. The value for type must be a valid J2EEManagedObject type. The valid types are:
 - J2EEDomain
 - J2EEServer
 - J2EEApplication
 - EJBModule
 - WebModule
 - ConnectorModule
 - EntityBean
 - StatefulSessionBean
 - StatelessSessionBean
 - MessageDrivenBean
 - Servlet
 - ResourceAdapter
 - JavaMail
 - JDBC
 - JDBCDataSource
 - JDBCDriver
 - JMS
 - JNDI
 - JTA
 - RMI_IIOP
 - URL
 - JVM
 - Node

- name - specifies the name of the J2EEManagedObject. (e.g. type=Servlet,name=My Cool Servlet). The value of the name property is arbitrary to the vendors implementation with the following exceptions:
 - The value of the name property for managed object instances of the J2EEDomain type must be equivalent to the domain name of the domain it manages.
 - The value of the name property for managed object instances of the Node type must be the host name of the primary interface of the corresponding host.
- <parent J2EEManagedObject type> - specifies parent-type properties which are derived from the valid managed object types defined above. The value of a parent-type property is equal to the value of the name property of the OBJECT_NAME of the parents of this managed object. These standard key properties provide a simple mechanism to define a hierarchy of managed object names which will be unique and identifiable. The standard key properties are defined as follows:

<parent J2EEManagedObject type>=<parent J2EEManagedObject name>

The required parent-type key properties for each managed object type are listed in Table JSR77.3-1.

Table JSR77.3-1 Required parent-type keys

Managed Object Type	required parent-type keys
J2EEDomain	none
J2EEServer	none
J2EEApplication	J2EEServer
EJBModule	J2EEServer
WebModule	J2EEServer
ConnectorModule	J2EEServer
EntityBean	EJBModule, J2EEServer
StatefulSessionBean	EJBModule, J2EEServer
StatelessSessionBean	EJBModule, J2EEServer
MessageDrivenBean	EJBModule, J2EEServer
Servlet	WebModule, J2EEServer

Table JSR77.3-1 Required parent-type keys

Managed Object Type	required parent-type keys
ResourceAdapter	ConnectorModule, J2EEServer
JavaMail	J2EEServer
JDBC	J2EEServer
JDBCDataSource	JDBC, J2EEServer
JDBCDriver	J2EEServer
JMS	J2EEServer
JNDI	J2EEServer
JTA	J2EEServer
RMI-IIOP	J2EEServer
URL	J2EEServer
JVM	J2EEServer
Node	J2EEServer

Suppose we have a J2EEDomain object whose name is:

```
FirstEverBank:type=J2EEDomain,name=FirstEverBank
```

When we create a J2EEServer its name could be as follows:

```
FirstEverBank:type=J2EEServer,name=Bank Server1
```

When we deploy a J2EEApplication into our server its name could be as follows:

```
FirstEverBank:type=J2EEApplication,name=AccountsController,J2EEServer=BankServer1
```

An EJBModule in our J2EEApplication could be named as follows:

```
FirstEverBank:type=EJBModule,name=BankAccount,J2EEServer=BankServer1
```

An Entity Bean in our EJBModule could be named as follows:

```
FirstEverBank:type=EntityBean,name=Account,EJBModule=BankAccount,J2EEServer=BankServer1
```


JSR77.3.1.1.2 stateManageable**boolean stateManageable**

If true, indicates that this managed object implements the StateManageable model and is state manageable by the specification of Chapter JSR77.5, “State Management”. If false, the managed object does not support state management.

JSR77.3.1.1.3 statisticsProvider**boolean statisticsProvider**

If true, indicates that the managed object supports the generation of performance statistics and implements the StatisticsProvider model (see “StatisticsProvider” on page 49).

If false, the J2EEManagedObject does not support performance statistics.

JSR77.3.1.1.4 eventProvider**boolean eventProvider**

If true, indicates that the managed object implements the EventProvider model (see “EventProvider” on page 41) and provides notification about events that occur on that object.

The semantic of the *eventProvider* attribute in the context of a J2EEDomain managed object is a special case. If its value is true, it indicates that the domain supports event notification and that the J2EEDomain managed object is a central registration point for event listeners that wish to receive event notification from all event providers in the domain. The J2EEDomain object must emit all events from all event providers in the managed object collection. In addition, the J2EEDomain managed object must support the emission of the *j2ee.object.created* and *j2ee.object.deleted* event types (see “type” on page 43). If any managed object in the domain is an event provider, i.e. the value of the managed object’s *eventProvider* attribute is true, then the J2EEDomain managed object must also be an event provider and the value of its *eventProvider* attribute must also be true. If the value of the *eventProvider* attribute of J2EEDomain managed object is false it indicates that none of the managed objects in the domain supports event notification.

JSR77.3.2 J2EEDomain extends J2EEManagedObject

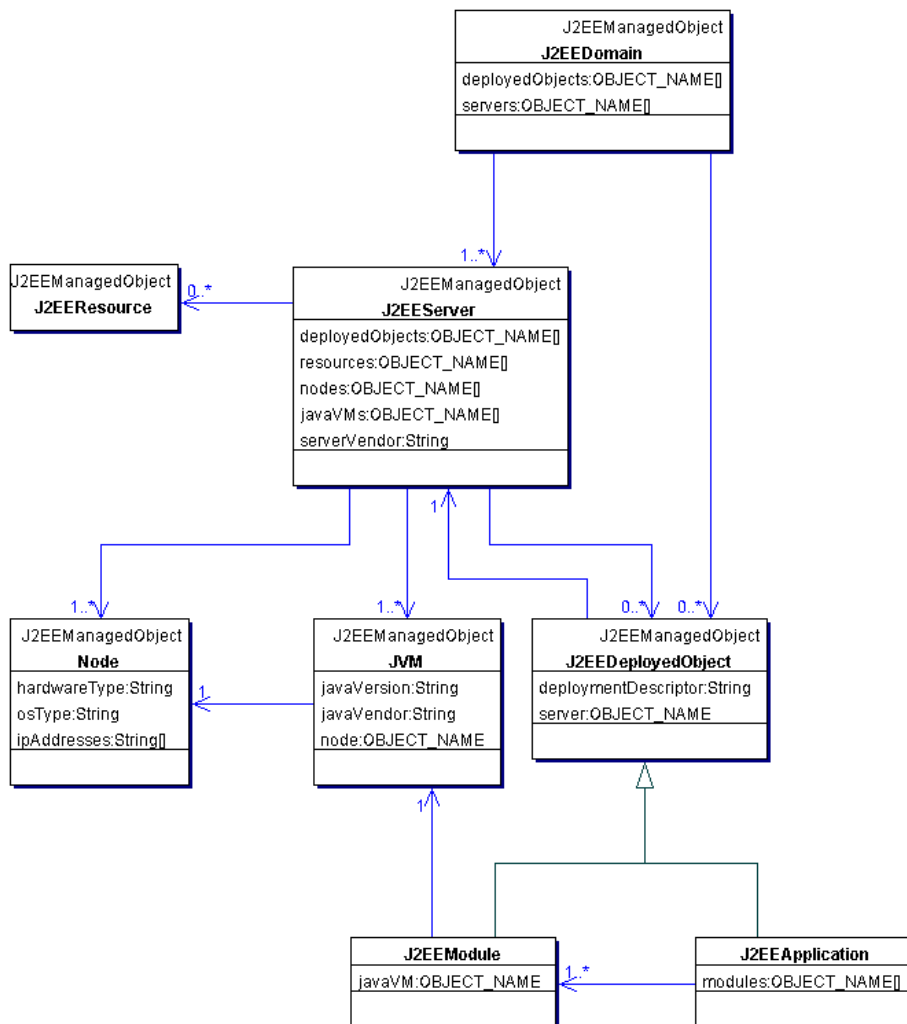


Figure JSR77.3.2 J2EEDomain

The J2EEDomain managed object type represents a management domain. There must be only one managed object that implements the J2EEDomain model per management domain. All servers and applications associated with the same domain are accessible from the J2EEDomain managed object.

JSR77.3.2.1 Attribute Detail***JSR77.3.2.1.1 deployedObjects*****OBJECT_NAME[] deployedObjects**

- supplier cardinality: 0..*

A list of all of the J2EEApplication and J2EEModule types deployed in this domain. For each application deployed in this domain, there must be one J2EEApplication managed object accessible from the *deployedObjects* list that identifies it. For each module deployed in this domain there must be one J2EEModule managed object accessible from the *deployedObjects* list that identifies it.

JSR77.3.2.1.2 servers**OBJECT_NAME[] servers**

- supplier cardinality: 1..*

A list of all J2EE Servers in this domain. For each J2EE Server running in the domain, there must be one J2EEServer managed object accessible from the *servers* list that identifies it.

JSR77.3.3 J2EEServer extends J2EEManagedObject

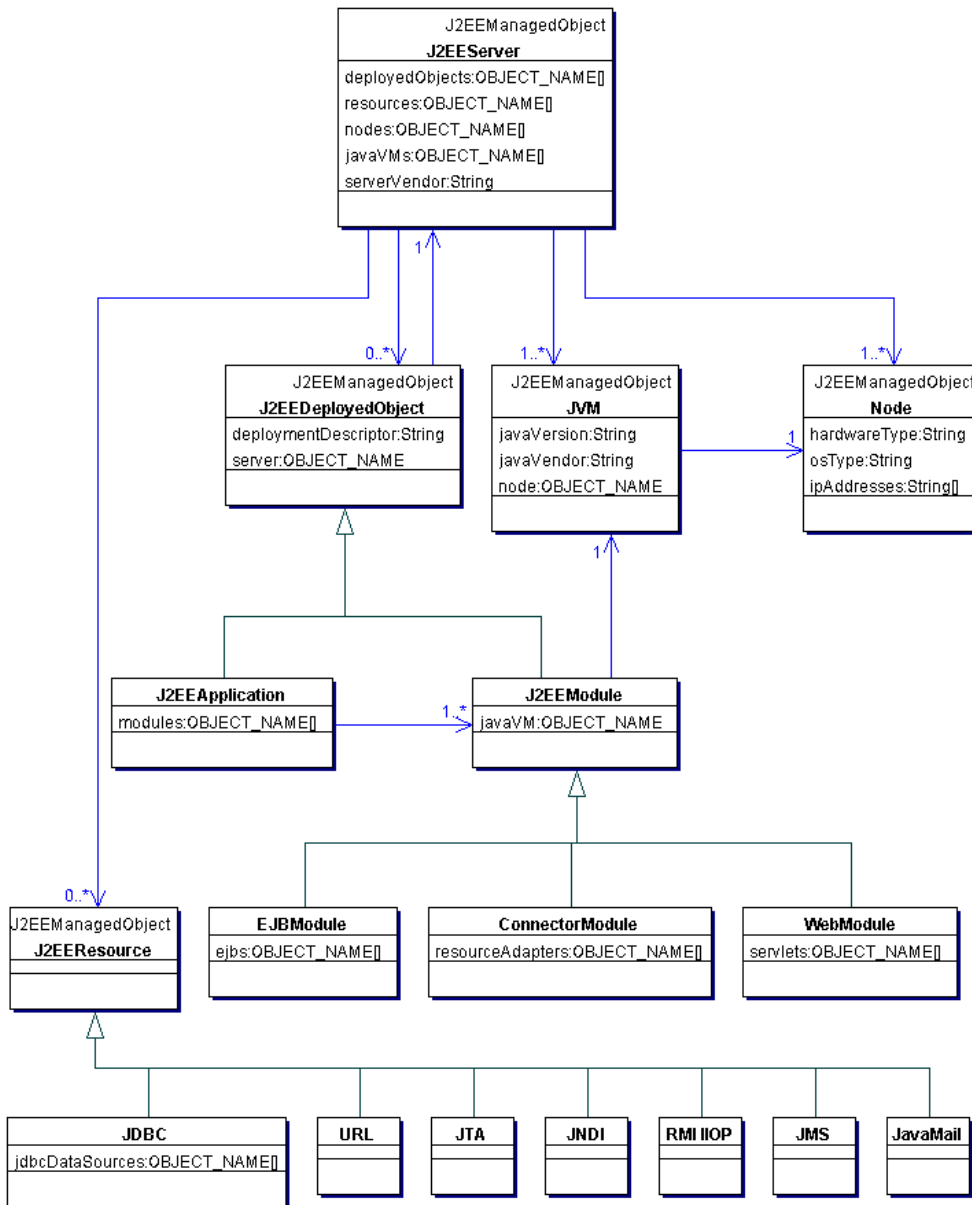


Figure JSR77.3.3 J2EEServer and associated managed objects

The J2EEServer model specifies the management information for a single J2EE server core implementation. The J2EE server core identifies the server core of one

instance of a J2EE platform product as described in the Java 2 Enterprise Edition Platform specification section 2.1, Architecture.

JSR77.3.3.1 Attribute Detail

JSR77.3.3.1.1 deployedObjects

OBJECT_NAME[] deployedObjects

- supplier cardinality: 0..*

A list of all of the J2EEApplication and J2EEModule types deployed on this J2EEServer. For each application deployed on this server, there must be one J2EEApplication managed object accessible from the *deployedObjects* list that identifies it. For each module deployed on this server, there must be one J2EEModule managed object accessible from the *deployedObjects* list that identifies it.

JSR77.3.3.1.2 resources

OBJECT_NAME[] resources

- supplier cardinality: 0..*

A list of resources available to this server. For each resource associated with this server, there must be one J2EEResource managed object accessible from the *resources* list that identifies it.

JSR77.3.3.1.3 nodes

OBJECT_NAME[] nodes

- supplier cardinality: 1..*

A list of nodes that this J2EEServer spans. For each machine that the server has components running on, there must be one Node managed object accessible from the *nodes* list that identifies it.

JSR77.3.3.1.4 *javaVMs*

OBJECT_NAME [] *javaVMs*

- supplier cardinality 1..*

A list of all Java virtual machines on which this J2EEServer has running threads. For each Java virtual machine this server utilizes, there must be one JVM managed object accesible from the *javaVMs* list that identifies it.

JSR77.3.3.1.5 *serverVendor*

String *serverVendor*

Identifies the J2EE platform vendor of this J2EEServer. The value of *serverVendor* is specified by the server vendor.

JSR77.3.4 JVM extends J2EEManagedObject

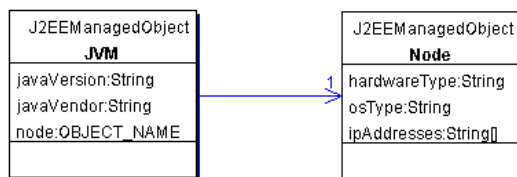


Figure JSR77.3.4 JVM type

Identifies a Java VM being utilized by a server.

JSR77.3.4.1 Attribute Detail

JSR77.3.4.1.1 *javaVersion*

String *javaVersion*

Identifies the Java Runtime Environment version of this Java VM. The value of *javaVersion* must be identical to the value of the system property *java.version*.

JSR77.3.4.1.2 javaVendor**String javaVendor**

Identifies the Java Runtime Environment vendor of this Java VM. The value of *javaVendor* must be identical to the value of the system property *java.vendor*.

JSR77.3.4.1.3 node**OBJECT_NAME node**

Identifies the node this JVM is running on.

JSR77.3.5 Node extends J2EEManagedObject

Identifies a hardware node or machine. The Node model specifies information about the machine that the Node represents. The name of the Node managed object must contain the primary hostname of the machine it represents.

JSR77.3.5.1 Attribute Detail**JSR77.3.5.1.1 hardwareType****String hardwareType**

The hardware type of this node.

JSR77.3.5.1.2 osType**String osType**

The operating system running on this node. The value of *osType* must be identical to the value of the system property *os.name*.

JSR77.3.5.1.3 ipAddresses**String [] ipAddresses**

Identifies the IPv4 or IPv6 IP addresses on this Node. For each IP address on a Node, there must be one *IpAddress* in the *ipAddresses* list that identifies it.

JSR77.3.6 J2EEDeployedObject extends J2EEManagedObject

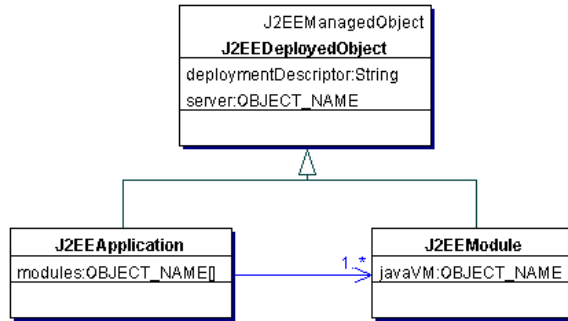


Figure JSR77.3.5 J2EEDeployedObject types

The base model for J2EEApplication and J2EEModule. All J2EEDeployedObject managed objects contain the original XML deployment descriptor that was created for the application or module during the deployment process.

JSR77.3.6.0.1 deploymentDescriptor

String deploymentDescriptor

The deploymentDescriptor string must contain the original XML deployment descriptor that was created for this module during the deployment process.

JSR77.3.6.0.2 server

OBJECT_NAME server

- supplier cardinality: 1

The J2EE server the application or module is deployed on.

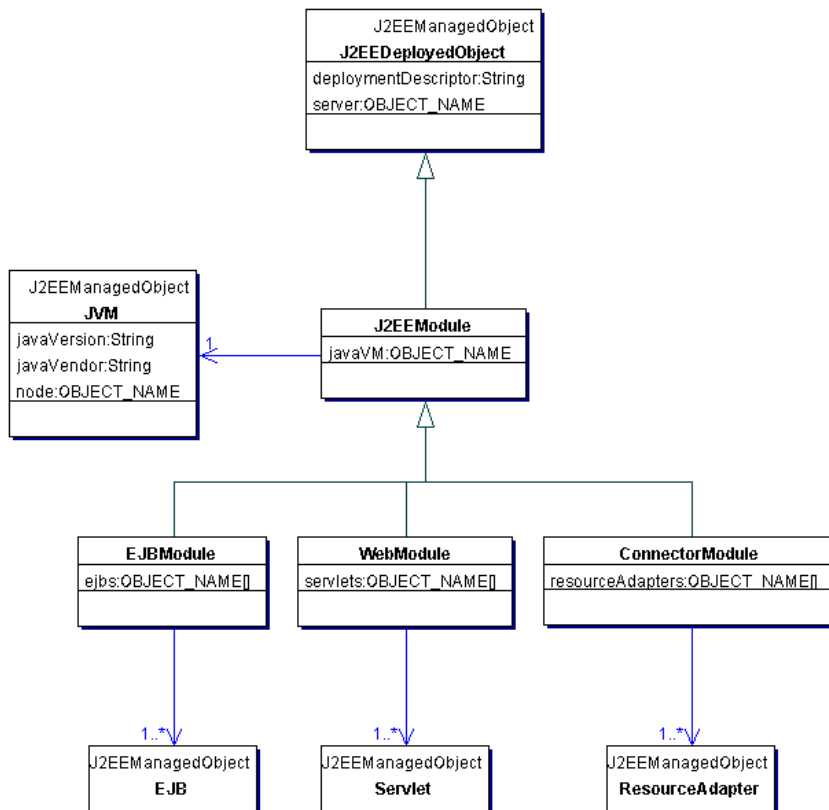
JSR77.3.7 J2EEApplication extends J2EEDeployedObject

Identifies a J2EE application EAR that has been deployed.

JSR77.3.7.1 Attribute Detail**JSR77.3.7.1.1 modules****OBJECT_NAME[] modules**

- supplier cardinality: 1..*

A list of J2EEModules that comprise this application. For each J2EE module that is utilized by this application, there must be one J2EEModule managed object accessible from the *modules* list that identifies it.

JSR77.3.8 J2EEModule extends J2EEDeployedObject**Figure JSR77.3.6** J2EEModule types

The J2EEModule model is the base model for all of the J2EE Module types. Managed objects that implement the J2EEModule model represent EAR, JAR, WAR and RAR files that have been deployed. Each J2EEModule may associated with one or more J2EEApplication objects with the exception of the ConnectorModule, which may or may not be associated with a J2EEApplication.

JSR77.3.8.1 Attribute Detail

JSR77.3.8.1.1 javaVM

OBJECT_NAME javaVM

Identifies the Java virtual machine on which this module is running.

JSR77.3.9 EJBModule extends J2EEModule

Identifies a deployed EJB JAR module.

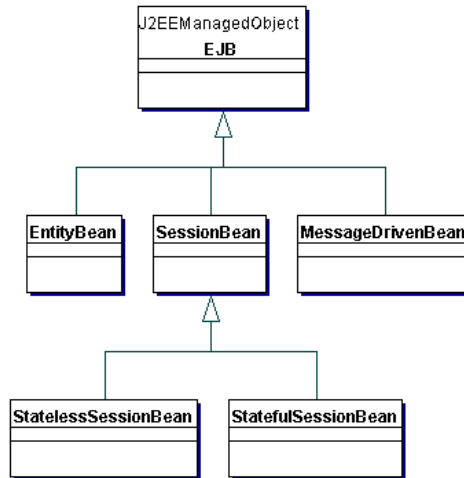
JSR77.3.9.1 Attribute Detail

JSR77.3.9.1.1 ejbs

OBJECT_NAME [] ejbs

- supplier cardinality: 1..*

A list of EJB components contained in the deployed EJB JAR module. For each EJB component contained in the deployed EJB JAR there must be one EJB managed object accessible from the *ejbs* list that identifies it.

JSR77.3.10 EJB extends J2EEManagedObject**Figure JSR77.3.7** EJB types

The EJB model is the base model for the specific types of Enterprise JavaBeans components.

JSR77.3.11 MessageDrivenBean extends EJB

Identifies an individual message-driven bean. For each deployed message-driven bean, there must be one managed object that implements the MessageDrivenBean model.

JSR77.3.12 EntityBean extends EJB

Identifies a deployed entity bean. For each deployed entity bean, there must be one managed object that implements the EntityBean model.

JSR77.3.13 SessionBean extends EJB

Identifies a deployed session bean. The SessionBean model is always extended by one of the sub-models StatefulSessionBean or StatelessSessionBean.

JSR77.3.14 StatefulSessionBean extends SessionBean

Identifies a deployed stateful session bean.

JSR77.3.15 StatelessSessionBean extends SessionBean

Identifies a deployed stateless session bean.

JSR77.3.16 WebModule extends J2EEModule

The WebModule model identifies a deployed WAR module.

JSR77.3.16.1 Attribute Detail**JSR77.3.16.1.1 servlets**

OBJECT_NAME [] servlets

- supplier cardinality: 1..*

A list of servlets contained in the deployed WAR module. For each servlet contained in the deployed WAR module there must be one Servlet managed object accessible from the *servlets* list that identifies it.

JSR77.3.17 Servlet extends J2EEManagedObject

Identifies a deployed servlet component.

JSR77.3.18 ConnectorModule extends J2EEModule

The ConnectorModule model identifies a deployed connector (RAR) module.

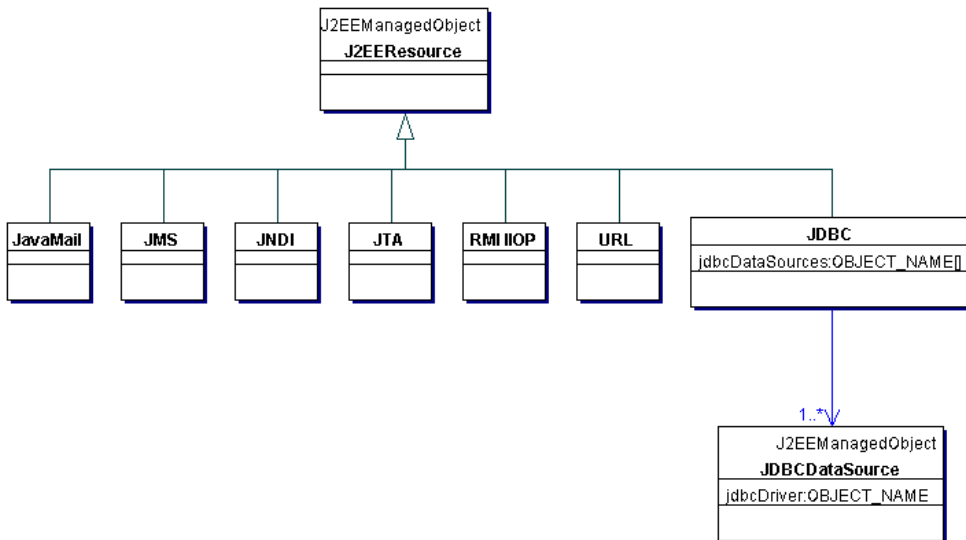
JSR77.3.18.1 Attribute Detail**JSR77.3.18.1.1 resourceAdapters****OBJECT_NAME [] resourceAdapters**

- supplier cardinality: 1..*

A list of resource adapters contained in this deployed connector (RAR) module. For each resource adapter contained in the deployed RAR module, there must be one ResourceAdapter managed object accessible from the *resourceAdapters* list that identifies it.

JSR77.3.19 ResourceAdapter extends J2EEManagedObject

Identifies a deployed resource adapter.

JSR77.3.20 J2EEResource extends J2EEManagedObject**Figure JSR77.3.8** J2EEResource

J2EEResource is the base model for all managed object types that represent J2EE resources. J2EE resources are resources utilized by the J2EE core server to provide the J2EE standard services required by the J2EE platform architecture. For each

J2EE standard service that a server provides, there must be one managed object that implements the J2EEResource model of the appropriate type.

JSR77.3.21 JavaMail extends J2EEResource

Identifies a JavaMail™ resource. For each JavaMail™ resource available to a server there must be one managed object that implements the JavaMail™ model.

JSR77.3.22 JMS extends J2EEResource

The JMS model identifies a Java Message Service resource. For each JMS resource provided on a server, there must be one managed object that implements the JMS model.

JSR77.3.23 JNDI extends J2EEResource

Identifies a Java Naming and Directory Interface™ (JNDI) resource. For each JNDI resource available to a server there must be one managed object that implements the JNDI model.

JSR77.3.24 JTA extends J2EEResource

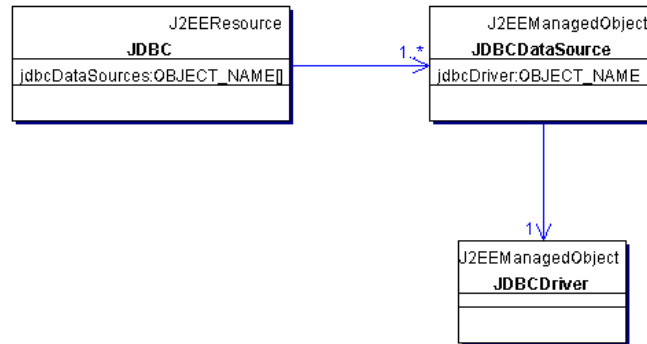
Identifies a Java Transaction API (JTA) resource. For each JTA resource available to a server there must be one managed object that implements the JTA model.

JSR77.3.25 RMI-IIOP extends J2EEResource

Identifies an RMI-IIOP resource. For each RMI-IIOP resource available to a server there must be one managed object that implements the RMI-IIOP model.

JSR77.3.26 URL extends J2EEResource

Identifies a URL resource. For each URL resource available to a server there must be one managed object that implements the URL model.

JSR77.3.27 JDBC extends J2EEResource**Figure JSR77.3.9** JDBC

The JDBC model identifies a JDBC resource. A JDBC resource manages one or more JDBC data sources. For each JDBC resource provided on a server, there must be one JDBC managed object accessible from the servers *resources* list that identifies it.

JSR77.3.27.1 Attribute Detail**JSR77.3.27.1.1 jdbcDataSource**

OBJECT_NAME[] jdbcDataSouces

- supplier cardinality: 1..*

Identifies the JDBC data sources available on the corresponding JDBC resource.

JSR77.3.28 JDBCDataSource extends J2EEManagedObject

Identifies a physical JDBC data source. For each JDBC data source available to a server there must be one managed object that implements the JDBCDataSource model.

JSR77.3.28.1 Attribute Detail

JSR77.3.28.1.1 jdbcDriver

OBJECT_NAME jdbcDriver

- supplier cardinality: 1

Identifies the JDBC driver for the corresponding JDBC data source.

JSR77.3.29 JDBCDriver extends J2EEManagedObject

Identifies an individual JDBC driver. For each JDBC driver available to a JDBC resource there must be one managed object that implements the JDBCDriver model.

CHAPTER JSR77.4

Events

This chapter specifies a standard extensible event model for the J2EE Management Model. Events notify management applications that need to react to state changes and specific conditions that occur in the platform containers and services.

JSR77.4.1 EventProvider

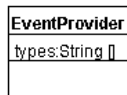


Figure JSR77.4.1 EventProvider

The EventProvider model specifies the types attribute, which must be included in all managed objects that emit events.

JSR77.4.1.1 Attribute Detail

JSR77.4.1.1.1 types

String [] types

A list of the types of events the managed object emits. The contents of the list are type strings as specified in “type” on page 43.

JSR77.4.2 Notification

Notification
source:OBJECT_NAME
timeStamp:long
sequenceNumber:long
type:String
message:String
userData:Object

Figure JSR77.4.2 Notification

The Notification model specifies the attributes that must be included in every event generated by the management system. A Notification includes the name of the source managed object, a timestamp, a sequence number, a string that identifies its type, a message that provides additional information about the event, and optional vendor specific user data. All management systems that support event notification must support all of the attributes of the Notification model. All events that are generated on behalf of managed objects that are part of or extend the J2EE Management Model must provide valid data for each of the attributes specified by the Notification model.

JSR77.4.2.1 Attribute Detail

JSR77.4.2.1.1 source

OBJECT_NAME source

The name of the managed object that generated this event. The source string must be equal to the name attribute of the managed object from which the event originated.

JSR77.4.2.1.2 timeStamp

long timeStamp

The time of the event represented as a long, whose value is the number of milliseconds since January 1, 1970, 00:00:00.

JSR77.4.2.1.3 sequence**long sequence**

The sequence number of the event. Identifies the position of the event in a stream of events. The sequence number provides a means of determining the order of sequential events that occurred with the same timestamp within the minimum supported unit of time. The scope of the sequence number is the source of the event.

JSR77.4.2.1.4 type**String type**

The type of an event, not to be confused with its Java class, is the characterization of a generic event object. The type is assigned by the source object and conveys the semantic meaning of a particular event. The type is given as a string which is interpreted as any number of dot-separated components. This allows some structure in the naming of event types.

All strings prefixed by “j2ee.” are reserved for the standard events defined by this specification. Otherwise, event sources are free to define all types they wish to use when naming the events they emit. All of the event types emitted by an event provider must be listed in the types attribute.

Standard event types are:

- j2ee.object.created- a new managed object was created.
- j2ee.object.deleted - a managed object was deleted.
- j2ee.state.starting - a state manageable object entered the starting state.
- j2ee.state.running - a state manageable object entered the running state.
- j2ee.state.stopping - a state manageable object entered the stopping state.
- j2ee.state.stopped - a state manageable object entered the stopped state.
- j2ee.state.failed - a state manageable object entered the failed state.
- j2ee.attribute.changed - an attribute has change value

JSR77.4.2.1.5 message

String message

An informational message about the event.

JSR77.4.2.1.6 userData

Object userData

Optional data the notification broadcaster wishes to communicate to listeners. The content of the data is vendor specific. The userData attribute may be null.

JSR77.4.3 Event Notification

Most management systems specify an event notification mechanism. Implementations of the J2EE Management Model by management systems that support event notification must deliver events that implement the Notification model.

Management tools that interoperate with these management systems and their protocols receive events through the delivery mechanisms specified by the management system being incorporated.

JSR77.4.3.1 javax.management.Notification

All implementations of this specification which deliver a Java object as an instance of an event notification, must generate event objects which are instances of the class `javax.management.Notification` as specified by the Java Management Extensions Instrumentation and Agent Specification (JSR003). The the value of the attributes in any instance of `javax.management.Notification` emitted by a managed object type specified by the J2EE Management Model must be exactly as specified in this chapter.

CHAPTER JSR77.5

State Management

This chapter specifies a standard state management model for the J2EE Management Model. State management refers to the management facilities that are provided by compliant J2EE platforms to manage the state of a J2EE platform and the components that comprise it. The management facilities allow Management Applications to get the current state of the platform and its components, find out how long the platform and components have been running, and start and stop the platform components.

JSR77.5.1 StateManageable

StateManageable
state:int startTime:long
start:void startRecursive:void stop:void

Figure JSR77.5.1 StateManageable

The StateManageable model specifies the operations and attributes that must be implemented by a managed object that supports state management. A managed object that implements the StateManageable model is termed a State Manageable Object (SMO). An SMO generates events when its state changes.

JSR77.5.1.1 Attribute Detail

JSR77.5.1.1.1 *state*

int state

The current state of this SMO.

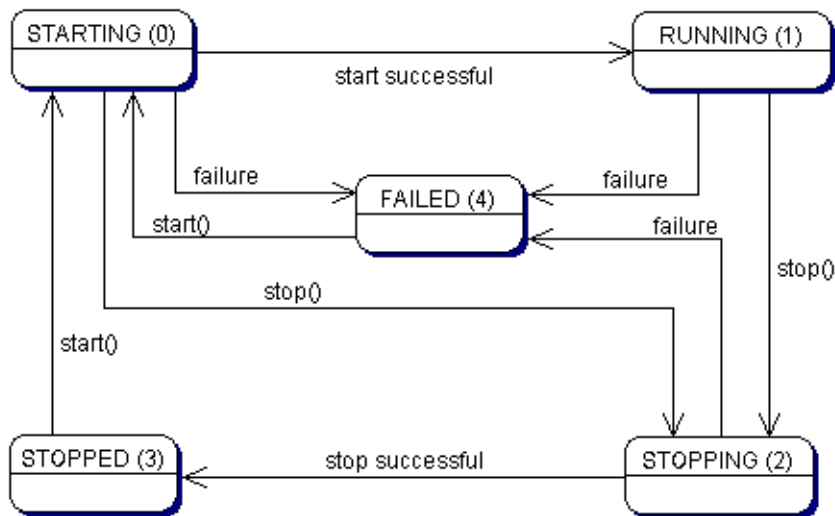


Figure JSR77.5.2 StateManageable object state diagram

The SMO can be in one of the following states (see the Event Model for details of the events):

STARTING (0)

This state indicates that the SMO has been requested to start, and is in the process of starting. On entering this state an SMO may generate an event whose type value is “j2ee.state.starting”. Event notification of the STARTING state is optional for all managed objects that implement StateManageable.

RUNNING (1)

This is the normal running state for an SMO. This state indicates that the SMO is operational. On entering this state an SMO must generate an event whose type value is “j2ee.state.running”. Event notification of the RUNNING state is required for all managed objects that implement StateManageable.

STOPPING (2)

This state indicates that the SMO has been requested to stop, and is in the process of stopping. On entering this state an SMO may generate an event whose type value is “j2ee.state.stopping”. Event notification of the STOPPING state is optional for all managed objects that implement StateManageable

STOPPED (3)

This state indicates that the StateManageable Object has stopped and can be restarted. On entering this state an SMO must generate an event whose type value is “j2ee.state.stopped”. Event notification of the STOPPED state is required by all managed objects that implement StateManageable.

FAILED (4)

This state indicates that the StateManageable Object has unexpectedly stopped. On entering this state an SMO must generate an event whose type value is “j2ee.state.failed”. Event notification of the FAILED state is required by all managed objects that implement StateManageable. It may be possible to start an SMO that has entered the FAILED state by calling the start() operation. It may be possible to stop an SMO that has entered the FAILED state by calling the stop() operation. An SMO that cannot successfully reach the RUNNING or STOPPED state from the FAILED state must reenter the FAILED state after the attempt to reach the RUNNING or STOPPED state fails and emit another “j2ee.state.failed” event. An SMO that cannot successfully reach the RUNNING or STOPPED state from the FAILED state requires intervention, such as redeployment, to restore the managed object to the desired state.

JSR77.5.1.1.2 *startTime*

long startTime

The time that the managed object was started (i.e. entered the RUNNING state) represented as a long, which value is the number of milliseconds since January 1, 1970, 00:00:00.

JSR77.5.1.2 Operation Detail**JSR77.5.1.2.1 *start***

void start()

Starts the SMO. This operation can be invoked only when the SMO is in the STOPPED state. It causes the SMO to go into the STARTING state initially, and if it completes successfully, the SMO will be in the RUNNING state. Note that start() is not called on any of the child SMOs that are registered with this SMO; it is the responsibility of the calling application to start the child SMO if this is required.

JSR77.5.1.2.2 startRecursive

void startRecursive()

Starts the SMO. This operation can only be invoked when the SMO is in the STOPPED state. It causes the SMO to go into the STARTING state initially, and if it completes successfully, the SMO will be in the RUNNING state. startRecursive() is called on all the child SMOs registered with this SMO that are in the STOPPED state.

JSR77.5.1.2.3 stop

void stop()

Stops the SMO. This operation can only be invoked when the SMO is in the RUNNING or STARTING state. It causes stop() to be called on all the child SMOs registered with this SMO that are in the RUNNING or STARTING state. It is mandatory if an SMO is in the STOPPED state, that all its child SMOs must also be in the STOPPED state, therefore there is no stopRecursive() operation. Invoking stop() causes the SMO to go into the STOPPING state initially, and if it completes successfully, the SMO and all the child SMOs will be in the STOPPED state.

CHAPTER JSR77.6

Performance Monitoring

This chapter describes the Performance Data Framework, which specifies a performance data model as well as performance data requirements of the J2EE Management Model.

JSR77.6.1 Performance Data Framework

The Performance Data Framework consists of the StatisticsProvider model, which any managed object may implement, the Stats interfaces, which specify standard performance attribute semantics for each managed object type, and the Statistic interfaces which provide specific interfaces for representing the common performance data types.

JSR77.6.2 StatisticsProvider

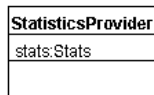


Figure JSR77.6.1 StatisticsProvider

The StatisticsProvider model must be implemented by all managed objects that provide performance data. A managed object that implements the StatisticsProvider model must have its *statisticProvider* attribute set to "true".

The *stats* attribute references the specific Stats interface that corresponds to the managed object type that is providing Statistics. For example, an EntityBean managed object that implements StatisticsProvider will have a reference in the *stats* attribute to an object that implements the EntityBeanStats interface. The detail for the *stats* attribute includes a table of the appropriate Stats interface that each managed object must implement if it provides performance data.

JSR77.6.2.1 Attribute Detail

JSR77.6.2.1.1 *stats*

Stats stats

Provides access to the implementation of the specific Stats interface that this managed object is required to support if it implements the StatisticProvider model. Table JSR77.6-1 shows the Stats interface required by each managed object type. See “Stats” on page 57 for details about support for specific statistics.

Table JSR77.6-1 Stats interface requirements

J2EEManagedObject	Required Stats interface
EJB	EJBStats
EntityBean	EntityBeanStats
JavaMail	JavaMailStats
JDBC	JDBCStats
JMS	JMSStats
JTA	JTAStats
JVM	JVMStats
MessageDrivenBean	MessageDrivenBeanStats
Servlet	ServletStats
SessionBean	SessionBeanStats
StatefulSessionBean	StatefulSessionBeanStats
StatelessSessionBean	StatelessSessionBeanStats
URL	URLStats

JSR77.6.3 package javax.management.j2ee.statistics

All of the following interfaces are part of the javax.management.j2ee.statistics package which provides the standard interfaces for Java implementations of the Statistic and Stats interfaces and subinterfaces. All implementations that provide statistics as Java object instances must implement the appropriate interfaces from the javax.management.j2ee.statistics package.

JSR77.6.4 Statistic

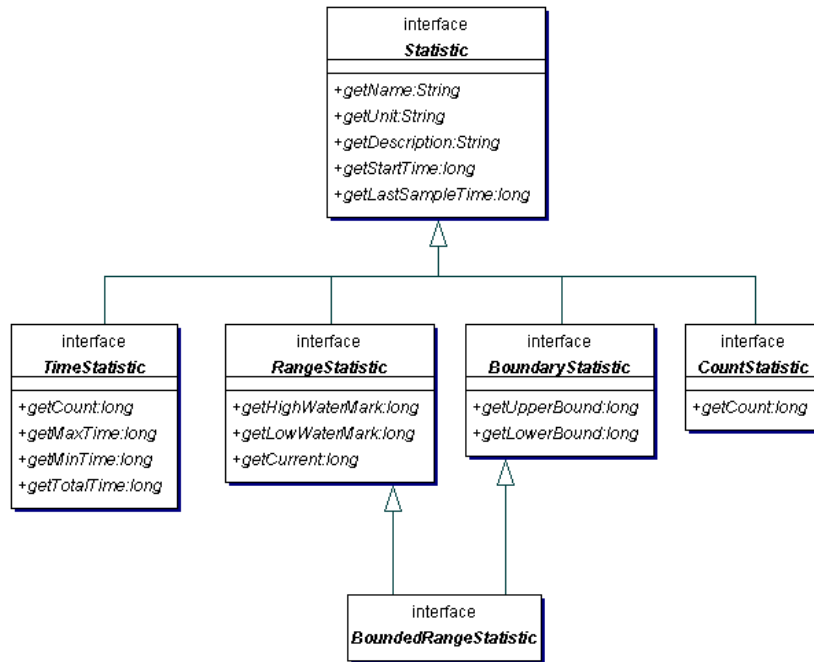


Figure JSR77.6.2 Statistic

The Statistic interface and its subinterfaces specify the required accessors which provide the performance data described by the specific attributes in the Stats interfaces. The Statistic subinterfaces specify accessors which provide statistical data about count, time, and both bounded and unbounded ranges.

JSR77.6.4.1 Operation Detail

JSR77.6.4.1.1 getName

String getName()

Returns the name of this Statistic. The name must always correspond to the name of the Stats accessor that is providing the data for this statistic.

JSR77.6.4.1.2 getUnit

String getUnit()

Returns the unit of measurement for this Statistic.

Valid values for TimeStatistic measurements are “**HOURL**”, “**MINUTE**”, “**SECOND**”, “**MILLISECOND**”, “**MICROSECOND**” and “**NANOSECOND**”.

JSR77.6.4.1.3 getDescription

String getDescription()

Returns a human-readable description of the Statistic.

JSR77.6.4.1.4 getStartTime

long getStartTime()

Returns the time the first measurment was taken represented as a long, whose value is the number of milliseconds since January 1, 1970, 00:00:00.

JSR77.6.4.1.5 getLastSampleTime

long getLastSampleTime()

Returns the time the most recent measurment was taken represented as a long, whose value is the number of milliseconds since January 1, 1970, 00:00:00.

JSR77.6.5 CountStatistic extends Statistic

The CountStatistic interface specifies standard count measurements.

JSR77.6.5.1 Operation Detail***JSR77.6.5.1.1 getCount*****long getCount()**

Returns the count since the measurement started.

JSR77.6.6 TimeStatistic extends Statistic

Specifies standard timing measurements for a given operation.

JSR77.6.6.1 Operation Detail***JSR77.6.6.1.1 getCount*****long getCount()**

Returns the number of times the operation was invoked since the beginning of this measurement.

JSR77.6.6.1.2 getMaxTime**long getMaxTime()**

Returns the maximum amount of time taken to complete one invocation of this operation since the beginning of this measurement.

JSR77.6.6.1.3 getMinTime**long getMinTime()**

Returns the minimum amount of time taken to complete one invocation of this operation since the beginning of this measurement.

JSR77.6.6.1.4 getTotalTime**long getTotalTime()**

Returns the sum total of time taken to complete every invocation of this operation since the beginning of this measurement. Dividing *totalTime* by *count* will give you the average execution time for this operation.

JSR77.6.7 RangeStatistic extends Statistic

Specifies standard measurements of the lowest and highest values an attribute has held as well as its current value.

JSR77.6.7.1 Operation Detail

JSR77.6.7.1.1 getHighWaterMark

long getHighWaterMark()

Returns the highest value this attribute has held since the beginning of the measurement.

JSR77.6.7.1.2 getLowWaterMark

long getLowWaterMark()

Returns the lowest value this attribute has held since the beginning of the measurement.

JSR77.6.7.1.3 getCurrent()

long getCurrent()

Returns the current value of this attribute.

JSR77.6.8 BoundaryStatistic extends Statistic

The BoundaryStatistic interface specifies standard measurements of the upper and lower limits of the value of an attribute.

JSR77.6.8.1 Operation Detail

JSR77.6.8.1.1 getUpperBound

long getUpperBound()

Returns the upper limit of the value of this attribute.

JSR77.6.8.1.2 getLowerBound

long getLowerBound()

Returns the lower limit of the value of this attribute.

JSR77.6.9 *BoundedRangeStatistic* extends *RangeStatistic*, *BoundaryStatistic*

The *BoundedRangeStatistic* interface extends the *RangeStatistic* and *BoundaryStatistic* interfaces and provides standard measurements of a range that has fixed limits.

JSR77.6.9.1 *Operation Detail*

JSR77.6.9.1.1 *getHighWaterMark*

long getHighWaterMark()

Returns the highest value this attribute has held since the beginning of the measurement. The value of *highWaterMark* must be less than or equal to *upperBound*.

JSR77.6.9.1.2 *getLowWaterMark*

long getLowWaterMark()

Returns the lowest value this attribute has held since the beginning of the measurement. The value of *lowWaterMark* must be greater than or equal to *lowerBound*.

JSR77.6.9.1.3 *getCurrent*

long getCurrent()

Returns the current value of this attribute.

JSR77.6.9.1.4 *getUpperBound*

long getUpperBound()

Returns the upper limit of the value of this attribute.

JSR77.6.9.1.5 *getLowerBound*

long getLowerBound()

Returns the lower limit of the value of this attribute.

JSR77.6.10 Stats

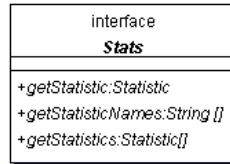


Figure JSR77.6.3 Stats hierarchy

The Stats interface and its subinterfaces specify performance data accessors for each of the specific managed object types. The data required by the interfaces is commonly available on all platform implementations. Managed objects that support statistics are permitted to provide support for a subset of the accessors in the Stats interfaces. Managed objects indicate which of the specified accessors return valid data by including only the names of the supported statistic accessors in the statisticNames list. The data provided by a supported statistic must be exactly as specified by the corresponding Stats interface. The Stats interfaces may be extended to provide vendor specific performance statistics. Vendor specific performance statistics must implement or extend one of the standard Statistics interfaces.

JSR77.6.10.1 Operation Detail

JSR77.6.10.1.1 getStatisticNames

String [] getStatisticNames()

Returns a list of the names of the attributes from the specific Stats interface that this object supports. Attributes named in the list must correspond to the names of operations in the Stats interface that will return Statistics object of the appropriate type which contains valid data. Each operation in a Stats interface is an accessor which follows the pattern *getAttributeName*. The *AttributeName* portion of the operation name is the value that is returned as the name in the StatisticNames list.

The value of attributes whose names are not included in the StatisticNames list is undefined and must be considered invalid. For each attribute name in the StatisticNames list that returns a Statistic there must be one Statistic object with the same name in the statistics list.

JSR77.6.10.1.2 getStatistics

Statistic [] getStatistics()

Returns a list of all the Statistic objects supported by this Stats object..

JSR77.6.10.1.3 getStatistic

Statistic getStatistic(String statisticName)

Gets a Statistic by name.

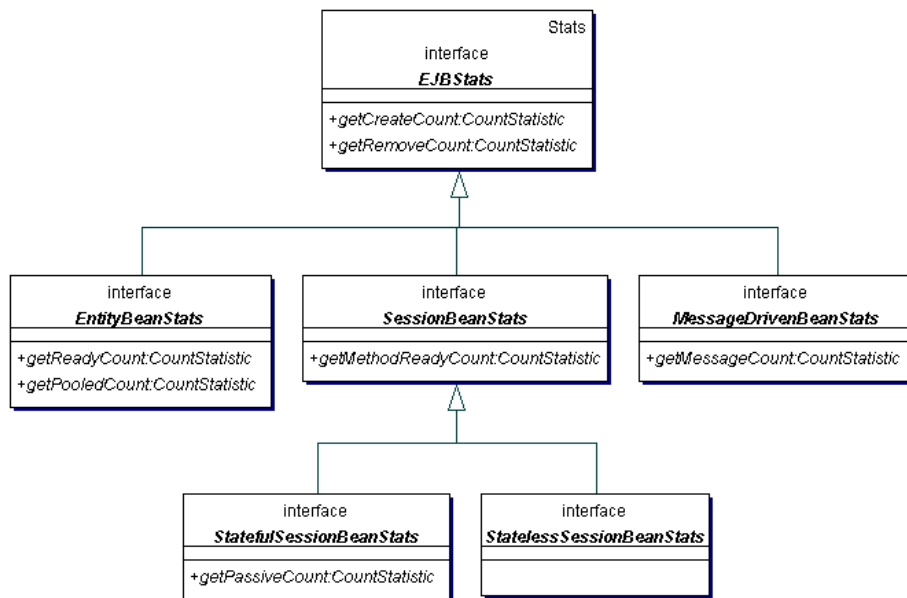
JSR77.6.11 EJBStats extends Stats

Figure JSR77.6.4 EJBStats

Specifies statistics provided by all EJB component types.

JSR77.6.11.1 Operation Detail**JSR77.6.11.1.1 getCreateCount**

CountStatistic getCreateCount()

Returns the number of times the beans create method was called.

JSR77.6.11.1.2 getRemoveCount

CountStatistic getRemoveCount()

Returns the number of times the beans remove method was called.

JSR77.6.12 EntityBeanStats extends EJBStats

Specifies statistics provided by entity beans.

JSR77.6.12.1 Operation Detail

JSR77.6.12.1.1 getReadyCount

CountStatistic getReadyCount()

Returns the number of bean instances in the ready state.

JSR77.6.12.1.2 getPooledCount

CountStatistic getPooledCount()

Returns the number of bean instances in the pooled state.

JSR77.6.13 MessageDrivenBeanStats extends EJBStats

Specifies the statistics provided by a message driven bean.

JSR77.6.13.1 Operation Detail

JSR77.6.13.1.1 getMessageCount

CountStatistic getMessageCount()

Returns the number of messages received.

JSR77.6.14 SessionBeanStats extends EJBStats

Specifies the statistics provided by session beans of both stateful and stateless types.

JSR77.6.14.1 Operation Detail***JSR77.6.14.1.1 getMethodReadyCount***

CountStatistic **getMethodReadyCount()**

Returns the number of beans in the method-ready state.

JSR77.6.15 StatefulSessionBeanStats extends SessionBeanStats

Specifies the statistics provided by a stateful session bean.

JSR77.6.15.1 Operation Detail***JSR77.6.15.1.1 getPassiveCount***

CountStatistic **getPassiveCount()**

Returns the number of beans that are in the passivated state.

JSR77.6.16 StatelessSessionBeanStats extends SessionBeanStats

Specifies the statistics provided by a stateless session bean.

JSR77.6.17 JavaMailStats extends Stats

interface
<i>JavaMailStats</i>
+ <i>getSentMailCount</i> :CountStatistic

Figure JSR77.6.5 JavaMailStats

Specifies the statistics provided by a JavaMail resource.

JSR77.6.17.1 Operation Detail***JSR77.6.17.1.1 getSentMailCount***

CountStatistic getSentMailCount()

Returns the number of mail messages sent.

JSR77.6.18 JDBCStats extends Stats

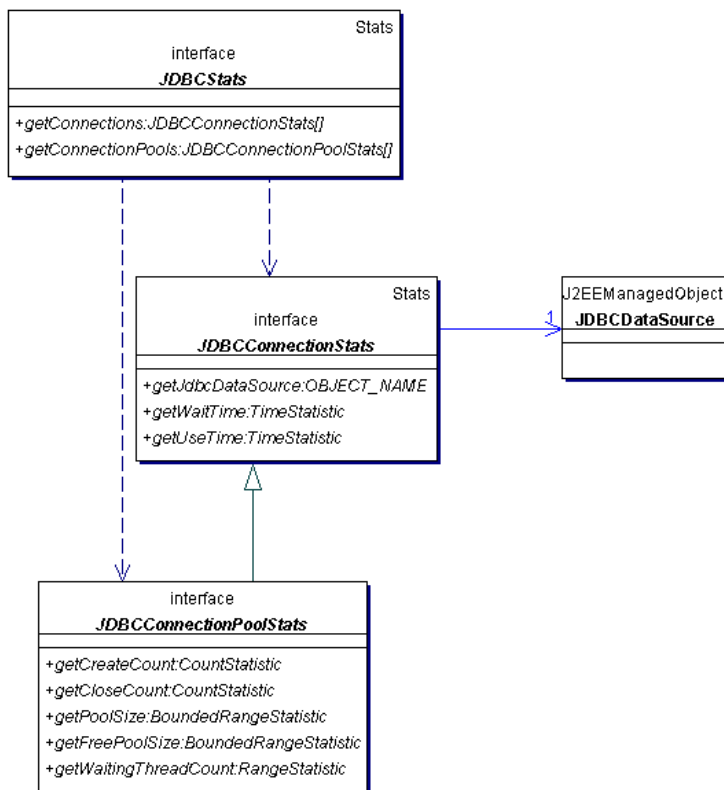


Figure JSR77.6.6 JDBCStats, JDBCConnectionStats, JDBCConnectionPoolStats

The JDBCStats type specifies the statistics provided by a JDBC resource.

JSR77.6.18.1 Operation Detail

JSR77.6.18.1.1 getConnections

JDBCConnectionStats[] getConnections

Returns a list of JDBCConnectionStats that provide statistics about the non-pooled connections associated with the referencing JDBC resource statistics.

JSR77.6.18.1.2 getConnectionPools

JDBCConnectionPoolStats[] getConnectionPools()

Returns a list of JDBCConnectionPoolStats that provide statistics about the connection pools associated with the referencing JDBC resource statistics.

JSR77.6.19 JDBCConnectionStats extends Stats

Specifies the statistics provided by a JDBC connection.

JSR77.6.19.1 Operation Detail

JSR77.6.19.1.1 getJdbcDataSource

OBJECT_NAME getJdbcDataSource()

Returns the name of the managed object that identifies the JDBC data source for this connection.

JSR77.6.19.1.2 getWaitTime

TimeStatistic getWaitTime()

Returns the time spent waiting for a connection to be available.

JSR77.6.19.1.3 getUseTime

TimeStatistic getUseTime()

Returns the time spent using a connection.

JSR77.6.20 JDBCConnectionPoolStats extends JDBCConnectionStats

Specifies the statistics provided by a JDBC connection pool.

JSR77.6.20.1 Operation Detail

JSR77.6.20.1.1 getCloseCount

CountStatistic getCloseCount()

Returns the number of connections closed.

JSR77.6.20.1.2 getCreateCount

CountStatistic getCreateCount()

Returns the number of connections created.

JSR77.6.20.1.3 getFreePoolSize

BoundedRangeStatistic getFreePoolSize()

Returns the number of free connections in the pool.

JSR77.6.20.1.4 getPoolSize

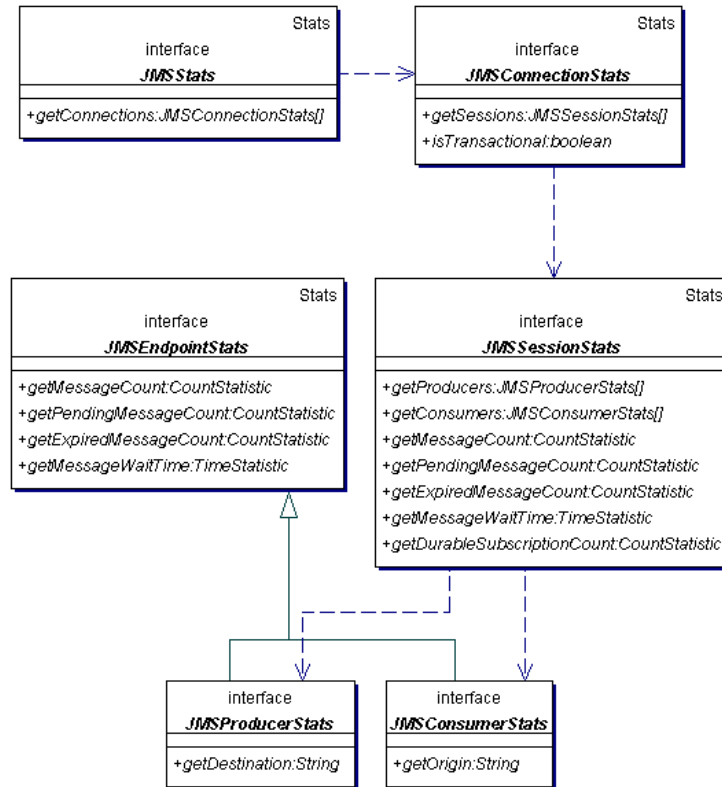
BoundedRangeStatistic getPoolSize()

Returns the size of the connection pool.

JSR77.6.20.1.5 getWaitingThreadCount

RangeStatistic getWaitingThreadCount()

Returns the number of threads waiting for a connection.

JSR77.6.21 JMSStats extends Stats**Figure JSR77.6.7** JMSStats, JMSConnectionStats, JMSSessionStats, JMSProducerStats, JMSConsumerStats

The JMSStats interface specifies the statistics provided by a JMS resource.

JSR77.6.21.1 Operation Detail**JSR77.6.21.1.1 getConnections**

JMSConnectionStats[] getConnections()

Returns a list of JMSConnectionStats that provide statistics about the connections associated with the referencing JMS resource.

JSR77.6.22 JMSConnectionStats extends Stats

Specifies the statistics provided by a JMS connection.

JSR77.6.22.1 Operation Detail

JSR77.6.22.1.1 getSession()

JMSSessionStats[] getSession()

Returns a list of JMSSessionStats that provide statistics about the sessions associated with the referencing JMSConnectionStats.

JSR77.6.22.1.2 isTransactional()

boolean isTransactional()

Returns the transactional state of this JMS connection. If true, indicates that this JMS connection is transactional.

JSR77.6.23 JMSSessionStats extends Stats

Specifies the statistics provided by a JMS session.

JSR77.6.23.1 Operation Detail

JSR77.6.23.1.1 getProducers()

JMSProducerStats[] getProducers()

Returns a list of JMSProducerStats that provide statistics about the message producers associated with the referencing JMS session statistics.

JSR77.6.23.1.2 getConsumers()

JMSConsumerStats[] getConsumers()

Returns a list of JMSConsumerStats that provide statistics about the message consumers associated with the referencing JMS session statistics.

JSR77.6.23.1.3 getDurableSubscriptionCount

CountStatistic `getDurableSubscriptionCount()`

Returns the number of durable subscriptions.

JSR77.6.23.1.4 getExpiredMessageCount

CountStatistic `getExpiredMessageCount()`

Returns the number of expired messages.

JSR77.6.23.1.5 getMessageCount

CountStatistic `getMessageCount()`

Returns the number of messages exchanged.

JSR77.6.23.1.6 getMessageWaitTime

TimeStatistic `getMessageWaitTime()`

Returns the time spent by a message before being delivered.

JSR77.6.23.1.7 getPendingMessageCount

CountStatistic `getPendingMessageCount()`

Returns the number of pending messages.

JSR77.6.24 JMSEndpointStats extends Stats

Specifies the base interface for the statistics provided by a JMS message producer or a JMS message consumer.

JSR77.6.24.1 Operation Detail

JSR77.6.24.1.1 getExpiredMessageCount

CountStatistic **getExpiredMessageCount()**

Returns the number of messages that expired before delivery.

JSR77.6.24.1.2 messageCount

CountStatistic **getMessageCount()**

Returns the number of messages sent or received.

JSR77.6.24.1.3 getMessageWaitTime

TimeStatistic **getMessageWaitTime()**

Returns the time spent by a message before being delivered.

JSR77.6.24.1.4 getPendingMessageCount

CountStatistic **getPendingMessageCount()**

Returns the number of pending messages.

JSR77.6.25 JMSProducerStats extends JMSEndpointStats

Specifies the statistics provided by a JMS message producer.

JSR77.6.25.1 Operation Detail

JSR77.6.25.1.1 getDestination

String getDestination()

Returns a string that encapsulates the identity of a message destination.

JSR77.6.26 JMSConsumerStats extends JMSEndpointStats

Specifies the statistics provided by a JMS message consumer.

JSR77.6.26.1 Operation Detail

JSR77.6.26.1.1 getOrigin

String getOrigin()

Returns a string that encapsulates the identity of a message origin.

JSR77.6.27 JTASStats extends Stats

interface		Stats
<i>JTASStats</i>		
<i>+getActiveCount:CountStatistic</i>		
<i>+getCommittedCount:CountStatistic</i>		
<i>+getRolledbackCount:CountStatistic</i>		

Figure JSR77.6.8 JTASStats

Specifies the statistics provided by a JTA resource.

JSR77.6.27.1 Operation Detail

JSR77.6.27.1.1 getActiveCount

CountStatistic **getActiveCount()**

Returns the number of active transactions.

JSR77.6.27.1.2 getCommittedCount

CountStatistic **getCommittedCount()**

Returns the number of committed transactions.

JSR77.6.27.1.3 getRolledbackCount

CountStatistic **getRolledbackCount()**

Returns the number of rolled-back transactions.

JSR77.6.28 JVMStats extends Stats

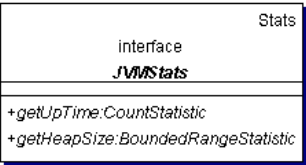


Figure JSR77.6.9 JVMStats

Specifies the statistics provided by a Java VM.

JSR77.6.28.1 Operation Detail

JSR77.6.28.1.1 getHeapSize

BoundedRangeStatistic getHeapSize()

Returns the size of the JVM's heap.

JSR77.6.28.1.2 getUpTime

CountStatistic getUpTime()

Returns the amount of time the JVM has been running.

JSR77.6.29 ServletStats extends Stats

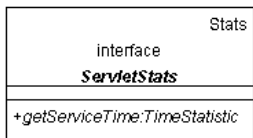


Figure JSR77.6.10 ServletStats

Specifies the statistics provided by a Servlet component.

JSR77.6.29.1 Operation Detail

JSR77.6.29.1.1 getServiceTime

TimeStatistic getServiceTime()

Returns the execution time of the servlets service method.

JSR77.6.30 URLStats extends Stats

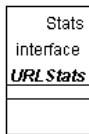


Figure JSR77.6.11 URLStats

Specifies the statistics provided by a URL resource.

JSR77.6.30.1 Operation Detail

There are no standard required URL Statistics defined by this specification. The URLStats interface may be extended to provide vendor specific performance statistics.

CHAPTER JSR77.7

J2EE Management EJB Component

The J2EE Management EJB component (MEJB) provides interoperable access to the J2EE Management Model from any J2EE component on all platforms that implement the J2EE Management specification. The MEJB component incorporates the standard framework for Java object instrumentation, the Java Management Extensions (JMX). The MEJB component exposes the managed objects on any J2EE platform as JMX manageable resources as defined by the Java Management Extensions Instrumentation and Agent Specification (JSR003). The MEJB component provides local and remote access of the platforms manageable resources through the EJB interoperability protocol.

JSR77.7.1 MEJB Implementation requirements

All implementations of this specification are required to provide access to the management model via a session bean which implements the `javax.management.j2ee.Management` interface as its remote interface.

All managed objects required by the J2EE Management Model must be available from the MEJB component.

All attributes required by the J2EE Management Model must be readable from the MEJB component and may be writable.

All operations required by the J2EE Management Model must be invocable from the MEJB component.

JSR77.7.1.1 JMX requirements

The MEJB component exposes the manageable resources on a J2EE platform as JMX Managed Beans (MBeans) and requires an implementation of the JMX public APIs specified by the Java Management Extensions Instrumentation and Agent Specification (JSR003).

JSR77.7.1.2 Remote Interface

The interface `javax.management.j2ee.Management` is the standard remote interface for the MEJB component. The remote interface of the MEJB component implementation must be `javax.management.j2ee.Management` or a subinterface of `javax.management.j2ee.Management`.

JSR77.7.1.3 Home Interface

The interface `javax.management.j2ee.ManagementHome` is the standard home interface for the MEJB component. The home interface of the MEJB component implementation must be `javax.management.j2ee.ManagementHome` or a subinterface of `javax.management.j2ee.ManagementHome`. The `create` method specified by the `ManagementHome` interface must return a session object that provides access to the entire managed object hierarchy.

JSR77.7.1.4 Naming Requirements

It is recommended that the home interface for the MEJB component be named in the `java:comp/env/mgmt` subcontext. It is recommended that a default name of `java:comp/env/mgmt/MEJB` be used when possible.

JSR77.7.1.5 Event Listener Requirements

Management Applications who wish to receive events from managed objects that support events must register an event listener object that implements the `javax.management.NotificationListener` interface specified in the Java Management Extensions Instrumentation and Agent Specification (JSR003).

It is not explicitly required that all event listeners communicate with remote objects, since internal listeners may operate in the same Java VM. In most cases the event listener registered by the management application will need to receive events from a remote object in order to interoperate with the managed system.

It is the responsibility of the implementation of the `ListenerRegistration` interface returned by the MEJB `getListenerRegistry()` method to implement an appropriate mechanism to receive events from the management system and deliver them to the event listener registered by the client. Figure JSR77.7.1 illustrates the event delivery paradigm. The listener connector and listener proxy are presented as conceptual types for the proprietary event delivery mechanism supported by an implementation of an MEJB. The transport mechanism for remote delivery is also implementation specific and may use services provided by the J2EE platform such as JMS or RMI.

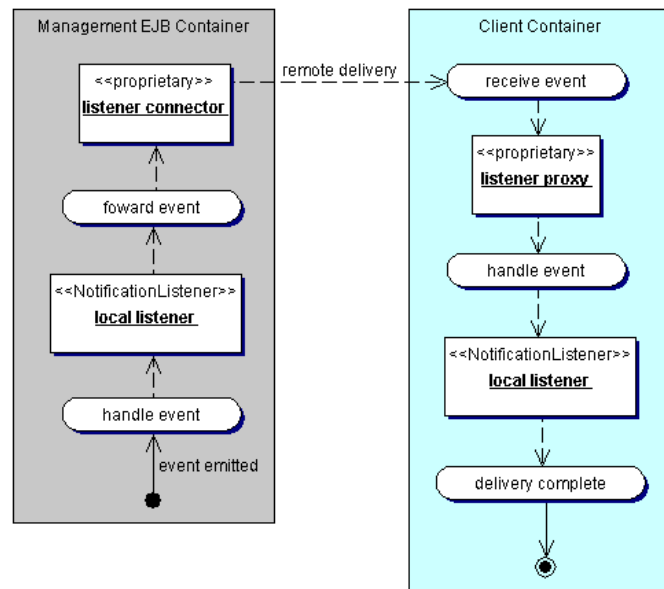


Figure JSR77.7.1 Event delivery

JSR77.7.2 Using the Management EJB component

This section provides code examples of a J2EE management component using the Management EJB component.

JSR77.7.2.1 Creating a MEJB Session Object

The code fragment in Code Example 7.2 illustrates how a J2EE component creates the MEJB session object.

```
Context ic = new InitialContext();
java.lang.Object objref = ic.lookup(ejbName);
ManagementHome home =
    (ManagementHome)PortableRemoteObject.narrow(objref,ManagementHome.class);
Management mejb = home.create();
```

Code Example 7.2 creating a MEJB session object from a J2EE application

JSR77.7.2.2 Finding managed objects of a certain type

The code fragment in Code Example 7.3 illustrates how the queryNames method can be used to find managed objects of a certain type.

```
// Find all J2EEApplication managed objects on this system
ObjectName searchpattern = new ObjectName("*:type=J2EEApplication,*");
Set managed_object_set = mejb.queryNames(searchpattern);
System.out.println("found " + managed_object_set.size() + " matching objects.");
```

Code Example 7.3 finding all J2EEApplication types

JSR77.7.2.3 Retrieving attribute information

The code fragment in Code Example 7.4 illustrates how the attributes of a managed object can be retrieved and their values determined.

```
Iterator managed_objects = managed_object_set.iterator();
while (managed_objects.hasNext()) {
    ObjectName objectname = (ObjectName)managed_objects.next();
    System.out.println(objectname);
    //get MBeanInfo and print the info
    MBeanInfo moi = mejb.getMBeanInfo(objectname);
    MBeanAttributeInfo[] atts = moi.getAttributes();
    for (int i = 0; i < atts.length; i++) {
        System.out.println("***** Attribute Info *****");
        System.out.println("Name " + atts[i].getName());
        System.out.println("Type " + atts[i].getType());
        System.out.println("isIs " + atts[i].isIs());
        System.out.println("isReadable " + atts[i].isReadable());
        System.out.println("isWritable " + atts[i].isWritable());
        System.out.println("value " + mejb.getAttribute(objectname, atts[i].getName()));
    }
}
```

Code Example 7.4 retrieving attribute information from a set of managed objects

JSR77.7.2.4 Invoking methods

The code fragment in Code Example 7.5 illustrates how a method with the signature `void test()` can be invoked.

```
mejb.invoke(appname, "test", new Object [] {}, new String [] {});
```

Code Example 7.5 invoking a method which takes no arguments

The code fragment in Code Example 7.6 illustrates how a method with the signature `String test(String, int)` can be invoked.

```
String sig1 = String.class.getName();  
String sig2 = Integer.TYPE.getName();  
String [] signature = new String [] {sig1, sig2};  
Object [] params = new Object [] {"This is a Test", new Integer(77)};  
String s = (String) mejb.invoke(appname, "test", params, signature);
```

Code Example 7.6 invoking a method with two arguments (String, int)

JSR77.7.2.5 Registering an event notification listener

The code fragment in Code Example 7.7 illustrates how the client registers an event listener.

```
// register an event listener  
objectname = new ObjectName(domainname + ":type=J2EEDomain,name=Hans' J2EE Domain");  
System.out.println("addNotificationListener( " + listener + " ) to " + objectname);  
ListenerRegistration lr = mejb.getListenerRegistry();  
lr.addNotificationListener(objectname, listener, null, "MEJBTester");
```

Code Example 7.7 registering an event listener

JSR77.7.3 package javax.management.j2ee

This section includes the complete API documentation for the J2EE Management EJB component interfaces.

All Interfaces Summary	
ListenerRegistration	Provides the methods to add and remove event listeners
Management	The remote interface for the MEJB
ManagementHome	The Home interface for the MEJB

JSR77.7.3.1 Interface *ListenerRegistration*

package: javax.management.j2ee

public class ListenerRegistration implements java.io.Serializable

Provides the methods to add and remove event listeners to the MEJB component.

Method Summary	
public void	addNotificationListener (javax.management.ObjectName name, javax.management.NotificationListener listener, javax.management.NotificationFilter filter, Object handback) Add a listener to a registered managed object.
public void	removeNotificationListener (javax.management.ObjectName name, javax.management.NotificationListener listener) Remove a listener from a managed object.

JSR77.7.3.1.1 Method Detail

addNotificationListener

```
public void addNotificationListener(javax.management.ObjectName name,  
    javax.management.NotificationListener listener, javax.management.  
    NotificationFilter filter, Object handback)
```

Add a listener to a registered managed object.

Throws:

javax.management.InstanceNotFoundException
java.rmi.RemoteException

Parameters:

name - The name of the managed object on which the listener should be added.
listener - The listener object which will handle the events emitted by the registered managed object.
filter - The filter object. If filter is null, no filtering will be performed before handling events.
handback - An opaque object to be sent back to the listener when a notification is emitted which helps the listener to associate information regarding the MBean emitter. This object cannot be used by the Notification broadcaster object. It should be resent unchanged with the notification to the listener.

Exception doc:

javax.management.InstanceNotFoundException - The managed object name provided does not match any of the registered managed objects.
java.rmi.RemoteException - a communication exception occurred during the execution of a remote method call.

removeNotificationListener

```
public void removeNotificationListener(javax.management.ObjectName name,  
    javax.management.NotificationListener listener)
```

Enables to remove a listener from a registered managed object.

Throws:

- javax.management.InstanceNotFoundException
- javax.management.ListenerNotFoundException
- java.rmi.RemoteException

Parameters:

name - The name of the managed object on which the listener should be removed.

listener - The listener object which will handle the events emitted by the registered managed object. This method will remove all the information related to this listener.

Exception doc:

javax.management.InstanceNotFoundException - The managed object name provided does not match any of the registered managed objects.

javax.management.ListenerNotFoundException - The listener is not registered in the managed object.

javax.rmi.RemoteException - a communication exception occurred during the execution of a remote method call.

JSR77.7.3.2 Interface *Management*

package: `javax.management.j2ee`

`public interface Management` extends `javax.ejb.EJBObject`

The Management interface provides the APIs to navigate and manipulate managed objects. The Management EJB component must implement this interface as its remote interface.

Method Summary	
public <code>Object</code>	<code>getAttribute</code> (<code>javax.management.ObjectName</code> name, <code>String</code> attribute) Gets the value of a specific attribute of a named managed object.
public <code>AttributeList</code>	<code>getAttributes</code> (<code>javax.management.ObjectName</code> name, <code>String[]</code> attributes) Enables the values of several attributes of a named managed object.
public <code>String</code>	<code>getDefaultDomain</code> () Returns the default domain name of this MEJB.
public <code>Integer</code>	<code>getMBeanCount</code> () Returns the number of managed objects registered in the MEJB.
public <code>MBeanInfo</code>	<code>getMBeanInfo</code> (<code>javax.management.ObjectName</code> name) Returns an object that allows the retrieval of the attributes and operations that a managed object exposes for management.
public <code>Object</code>	<code>invoke</code> (<code>javax.management.ObjectName</code> name, <code>String</code> operationName, <code>Object[]</code> params, <code>String[]</code> signature) Invokes an operation on a managed object.
public <code>boolean</code>	<code>isRegistered</code> (<code>javax.management.ObjectName</code> name) Checks whether a managed object, identified by its object name, is already registered with the MEJB.
public <code>Set</code>	<code>queryNames</code> (<code>javax.management.ObjectName</code> name, <code>javax.management.QueryExp</code> query) Gets the names of managed objects controlled by the MEJB.
public <code>void</code>	<code>setAttribute</code> (<code>javax.management.ObjectName</code> name, <code>javax.management.Attribute</code> attribute) Sets the value of a specific attribute of a named managed object.
public <code>AttributeList</code>	<code>setAttributes</code> (<code>javax.management.ObjectName</code> name, <code>javax.management.AttributeList</code> attributes) Sets the values of several attributes of a named managed object.

public ListenerRegistration	getListenerRegistry() Returns the ListenerRegistration implementation for this MEJB component.
---	--

JSR77.7.3.2.1 Method Detail**getAttribute**

```
public Object getAttribute(javax.management.ObjectName name, String attribute)
```

Gets the value of a specific attribute of a named managed object. The managed object is identified by its object name.

Throws:

[javax.management.MBeanException](#)
[javax.management.AttributeNotFoundException](#)
[javax.management.InstanceNotFoundException](#)
[javax.management.ReflectionException](#)
[java.rmi.RemoteException](#)

Parameters:

name - The object name of the managed object from which the attribute is to be retrieved.
 attribute - A String specifying the name of the attribute to be retrieved.

Returns:

The value of the retrieved attribute.

Exception doc:

[javax.management.AttributeNotFoundException](#) - The attribute specified is not accessible in the managed object.
[javax.management.MBeanException](#) - Wraps an exception thrown by the managed object's getter.
[javax.management.InstanceNotFoundException](#) - The managed object specified is not registered in the MEJB.
[javax.management.ReflectionException](#) - Wraps a [java.lang.Exception](#) thrown when trying to invoke the setter.
[javax.rmi.RemoteException](#) - a communication exception occurred during the execution of a remote method call.

getAttributes

```
public javax.management.AttributeList getAttributes(javax.management.ObjectName name, String[] attributes)
```

Gets the values of several attributes of a named managed object. The managed object is identified by its object name.

Throws:

- javax.management.InstanceNotFoundException
- javax.management.ReflectionException
- java.rmi.RemoteException

Parameters:

- name - The object name of the managed object from which the attributes are retrieved.
- attributes - A list of the attributes to be retrieved.

Returns:

An instance of javax.management.AttributeList which contains a list of the retrieved attributes as javax.management.Attribute instances.

Exception doc:

- javax.management.InstanceNotFoundException - The managed object specified is not registered in the MEJB.
- javax.management.ReflectionException - An exception occurred when trying to invoke the getAttributes method of a Dynamic managed object.
- java.rmi.RemoteException - a communication exception occurred during the execution of a remote method call.

getDefaultDomain

```
public String getDefaultDomain()
```

Returns the default domain name of this MEJB.

Throws:

- java.rmi.RemoteException

Exception Doc:

- java.rmi.RemoteException - a communication exception occurred during the execution of a remote method call.

getMBeanCount

```
public Integer getMBeanCount()
```

Returns the number of managed objects registered in the MEJB.

Throws:

java.rmi.RemoteException

Exception doc:

javax.rmi.RemoteException - a communication exception occurred during the execution of a remote method call.

getMBeanInfo

```
public javax.management.MBeanInfo getMBeanInfo(javax.management.Object-  
Name name)
```

This method discovers the attributes and operations that a managed object exposes for management.

Throws:

javax.management.IntrospectionException

javax.management.InstanceNotFoundException

javax.management.ReflectionException

java.rmi.RemoteException

Parameters:

name - The object name of the managed object to analyze.

Returns:

An instance of `javax.management.MBeanInfo` allowing the retrieval of all attributes and operations of this managed object.

Exception doc:

javax.management.IntrospectionException An exception occurred during introspection.

javax.management.InstanceNotFoundException The managed object specified was not found.

javax.management.ReflectionException An exception occurred when trying to perform reflection on the managed object.

java.rmi.RemoteException - a communication exception occurred during the execution of a remote method call.

invoke

```
public Object invoke(javax.management.ObjectName name, String operation-  
Name, Object[] params, String[] signature)
```

Invokes an operation on a managed object.

Throws:

- javax.management.InstanceNotFoundException
- javax.management.MBeanException
- javax.management.ReflectionException
- java.rmi.RemoteException

Parameters:

- name - The object name of the managed object on which the method is to be invoked.
- operationName The name of the operation to be invoked.
- params - An array containing the parameters to be set when the operation is invoked
- signature - An array containing the signature of the operation. Each element of the array contains a fully-qualified name of the entity (class, interface, array class, primitive type) that corresponds with a parameter type in the method's signature. The format of the strings must be as specified by `java.lang.Class.getName()`. The class objects will be loaded using the same class loader as the one used for loading the managed object on which the operation was invoked.

Returns:

The object returned by the operation, which represents the result of invoking the operation on the managed object specified.

Exception doc:

- javax.management.InstanceNotFoundException - The managed object specified is not registered in the MEJB.
- javax.management.MBeanException - Wraps an exception thrown by the managed object's invoked method.
- javax.management.ReflectionException - Wraps a `java.lang.Exception` thrown while trying to invoke the method.
- java.rmi.RemoteException - a communication exception occurred during the execution of a remote method call.

isRegistered

public boolean isRegistered(javax.management.ObjectName name)

Checks whether a managed object, identified by its object name, is already registered with the MEJB.

Throws:

java.rmi.RemoteException

Parameters:

name - The object name of the managed object to be checked.

Returns:

True if the managed object is already registered in the MEJB, false otherwise.

Exception doc:

javax.rmi.RemoteException - a communication exception occurred during the execution of a remote method call.

queryNames

public java.util.Set queryNames(javax.management.ObjectName name, javax.management.QueryExp query)

Gets the names of managed objects controlled by the MEJB. This method enables any of the following to be obtained: The names of all managed objects, the names of a set of managed objects specified by pattern matching on the ObjectName and/or a query expression, a specific managed object name (equivalent to testing whether a managed object is registered). When the object name is null or no domain and key properties are specified, all objects are selected. It returns the set of ObjectNames for the managed objects selected.

Throws:

java.rmi.RemoteException

Parameters:

name - The object name pattern identifying the managed objects to be retrieved. If null or no domain and key properties are specified, all the managed objects registered will be retrieved.

query - The query expression to be applied for selecting managed objects. If null no query expression will be applied for selecting managed objects.

Returns:

A set containing the ObjectNames for the managed objects selected. If no managed object satisfies the query, an empty set is returned.

Exception doc:

javax.rmi.RemoteException - a communication exception occurred during the execution of a remote method call.

setAttribute

```
public void setAttribute(javax.management.ObjectName name, javax.management.Attribute attribute)
```

Sets the value of a specific attribute of a named managed object. The managed object is identified by its object name.

Throws:

javax.management.InstanceNotFoundException
 javax.management.AttributeNotFoundException
 javax.management.InvalidAttributeValueException
 javax.management.MBeanException
 javax.management.ReflectionException
 java.rmi.RemoteException

Parameters:

name - The name of the managed object within which the attribute is to be set.
 attribute - The identification of the attribute to be set and the value it is to be set to.

Returns:

The value of the attribute that has been set.

Exception doc:

javax.management.InstanceNotFoundException - The managed object specified is not registered in the MEJB.
 javax.management.AttributeNotFoundException - The attribute specified is not accessible in the managed object.
 javax.management.InvalidAttributeValueException - The value specified for the attribute is not valid.
 javax.management.MBeanException - Wraps an exception thrown by the managed object's setter.
 javax.management.ReflectionException - Wraps a java.lang.Exception thrown when trying to invoke the setter.
 javax.rmi.RemoteException - a communication exception occurred during the execution of a remote method call.

setAttributes

```
public javax.management.AttributeList setAttributes(javax.management.ObjectName name, javax.management.AttributeList attributes)
```

Sets the values of several attributes of a named managed object. The managed object is identified by its object name.

Throws:

- javax.management.InstanceNotFoundException
- javax.management.ReflectionException
- java.rmi.RemoteException

Parameters:

- name - The object name of the managed object within which the attributes are to be set.
- attributes - A list of attributes: The identification of the attributes to be set and the values they are to be set to.

Returns:

The list of attributes that were set, with their new values.

Exception doc:

- javax.management.InstanceNotFoundException - The managed object specified is not registered in the MEJB.
- javax.management.ReflectionException - An exception occurred when trying to invoke the getAttributes method of a Dynamic managed object.
- javax.rmi.RemoteException - a communication exception occurred during the execution of a remote method call.

getListenerRegistry

```
public ListenerRegistration getListenerRegistry()
```

Returns the ListenerRegistration implementation for the MEJB component implementation which allows the client to register a event notification listener.

Throws:

- java.rmi.RemoteException

Returns:

An implementation of ListenerRegistration.

Exception doc:

- javax.rmi.RemoteException - a communication exception occurred during the execution of a remote method call.

JSR77.7.3.3 Interface *ManagementHome*

package: javax.management.j2ee

public interface *ManagementHome* extends javax.ejb.EJBHome

The required home interface for the Management EJB component. The interface may be extended by a proprietary interface to include additional create methods that take initialization arguments. A J2EE client must be able to create a compliant session object using the specified create method which takes no arguments.

Method Summary	
public void	create() Create an MEJB session object.

JSR77.7.3.3.1 Method Detail

create

public void create()

Throws:

javax.ejb.CreateException
java.rmi.RemoteException

Creates an MEJB session object which provides access to the J2EE Management Model.

Exception doc:

javax.ejb.CreateException - indicates a failure to create an EJB Object.
java.rmi.RemoteException - a communication exception occurred during the execution of a remote method call.

CHAPTER JSR77.8

J2EE Management SNMP

Note – This chapter is the starting point for an Internet-Draft. Internet-Drafts are working documents of the Internet Engineering Task Force (IETF), its areas, and its working groups. To facilitate initial rounds of review, it has been incorporated into this specification. It is anticipated that after completion of Public Review, this chapter will be replaced by a reference to the separate, formal Internet-Draft based on this chapter.

JSR77.8.1 Abstract

This chapter defines a portion of the Management Information Base (MIB) for use with network and systems management protocols in the Internet community. More specifically, it describes a set of managed objects that allow for the monitoring of the Java 2 Enterprise Edition Platform.

JSR77.8.2 Introduction

As part of the Java Community Process (see <http://www.jcp.org/>), JSR 77 was formed to define a management information model for the Java 2 Enterprise Edition Platform, the J2EE Management Model. The J2EE Management Model is a UML model which describes the structure, format, and semantics of management data for the J2EE Platform.

This chapter defines a portion of the Management Information Base (MIB) for use with network management protocols in the Internet community which is

consistent with that model as well as with the Structure of Management Information defined for the SNMP Management Framework.

The key words “MUST”, “MUST NOT”, “REQUIRED”, “SHALL”, “SHALL NOT”, “SHOULD”, “SHOULD NOT”, “RECOMMENDED”, “MAY”, and “OPTIONAL” in this document are to be interpreted as described in RFC 2119 [RFC2119].

JSR77.8.3 The SNMP Management Framework

The SNMP Management Framework presently consists of five major components:

- An overall architecture, described in RFC 2571 [RFC2571].
- Mechanisms for describing and naming objects and events for the purpose of management. The first version of this Structure of Management Information (SMI) is called SMIV1 and described in STD 16, RFC 1155 [RFC1155], STD 16, RFC 1212 [RFC1212] and RFC 1215 [RFC1215]. The second version, called SMIV2, is described in STD 58, RFC 2578 [RFC2578], STD 58, RFC 2579 [RFC2579] and STD 58, RFC 2580 [RFC2580].
- Message protocols for transferring management information. The first version of the SNMP message protocol is called SNMPv1 and described in STD 15, RFC 1157 [RFC1157]. A second version of the SNMP message protocol, which is not an Internet standards track protocol, is called SNMPv2c and described in RFC 1901 [RFC1901] and RFC 1906 [RFC1906]. The third version of the message protocol is called SNMPv3 and described in RFC 1906 [RFC1906], RFC 2572 [RFC2572] and RFC 2574 [RFC2574].
- Protocol operations for accessing management information. The first set of protocol operations and associated PDU formats is described in STD 15, RFC 1157 [RFC1157]. A second set of protocol operations and associated PDU formats is described in RFC 1905 [RFC1905].
- A set of fundamental applications described in RFC 2573 [RFC2573] and the view-based access control mechanism described in RFC 2575 [RFC2575].

A more detailed introduction to the current SNMPv3 Management Framework can be found in RFC 2570 [RFC2570]. A framework, termed the AgentX framework, which extends the SNMP Management Framework and can

accommodate environments wherein multiple components might need to have a shared mechanism for sending and receiving SNMP protocol messages, is described in RFC 2741 [RFC2741].

In SNMP, managed objects are accessed via a virtual information store, termed the Management Information Base or MIB. Objects in the MIB are defined using the mechanisms defined in the SMI.

This chapter specifies a MIB module that is compliant to the SMIV2. A MIB conforming to the SMIV1 can be produced through the appropriate translations. Some machine readable information in SMIV2 will be converted into textual descriptions in SMIV1 during the translation process. However, this loss of machine readable information is not considered to change the semantics of the MIB.

JSR77.8.4 Overview

The J2EE MIB module defined in this chapter can be used to discover and monitor details about J2EE servers. The status of the server and its components is reported as MIB objects which can be polled, and MAY also be reported as traps. OPTIONAL performance statistics are also defined.

The MIB consists of the following tables:

- Domains (j2eeDomTable)
- Servers (j2eeSrvTable)
- Nodes (j2eeNodeTable)
- JVMs (j2eeJVMTTable)
- Applications (j2eeAppTable)
- Modules (j2eeModTable)
- Enterprise Java Beans (j2eeBeanTable)
- Servlets (j2eeSletTable)
- Resource Adapters (j2eeAdapTable)
- Resources (j2eeRsrcTable)

- JDBC Data Drivers and Sources (j2eeJDBCTable)

The J2EE management model allows multiple servers on a single node as well as single servers spanning multiple nodes. The MIB is defined and indexed in support of this, and permits many but not all alternatives in agent and instrumentation architectures; these MAY be based on either monolithic or AgentX-like SNMP frameworks. All of the following are supported:

- Components of different multi-node servers may coexist on a node, from one or more vendors.
- The agent (or AgentX master agent, or, less likely, the AgentX subagent) might reside on a node which does not host any component of a server.
- A J2EE management domain (part of the J2EE management model) can span multiple servers and therefore multiple nodes.
- Servers and domains may share nodes with other SNMP-managed components.
- An AgentX-like master agent might be present, with sessions to one or more subagents having access to a server's instrumentation.

An SNMP agent (or agentx-like subagent) supporting this MIB MUST assemble and present an integrated view of any server, whether single node or multi- node, for which it has created a row in the server table (j2eeSrvTable).

JSR77.8.5 The Structure of the MIB

The following guidelines were followed for the MIB:

- Managed object types from the model become conceptual tables;
- Some attribute lists for certain managed object types also become conceptual tables;
- Attributes inherited from supertypes become conceptual columns in tables for all managed object types inheriting that attribute;
- Key property lists used to assign unique names to managed objects become the indices into a conceptual table entry;

- Events are mapped into traps; o State management is embedded into the individual tables;
- the Performance Data Framework become augmentations of conceptual tables (or were embedded into the tables directly in cases where the table was specific to a particular managed object); those augmentations (or their embedded equivalents) are organized into object groups which are referenced in module compliance macros in the definition of what subsets of the MIB may be implemented while still conforming to this definition.
- Truth values indicate on a per-table basis at runtime whether an optional feature (events, state management, performance statistics) is supported.
- Only a subset of J2EE management events are issued as traps. No creation, deletion, or attribute changes are reported as traps. Only “core” (see J2eeSMState) state changes are issued, and only for servers, resources, entity beans, and message-driven beans.

Note that all strings defined in the MIB are up to 255 characters in length; no such limitation has been specified in the attributes for the UML model.

Note that it is considered out-of-scope for this MIB to include objects which could facilitate the discovery of an authoritative source for MIB information about multi-node servers. Such a mapping is considered an SNMP client concern.

JSR77.8.6 Definitions

J2EE-MIB DEFINITIONS ::= BEGIN

IMPORTS

MODULE-IDENTITY, OBJECT-TYPE, NOTIFICATION-TYPE,
Integer32, Unsigned32, enterprises
FROM SNMPv2-SMI

TruthValue, DateAndTime, TAddress
FROM SNMPv2-TC

MODULE-COMPLIANCE, OBJECT-GROUP, NOTIFICATION-GROUP
FROM SNMPv2-CONF

SnmpAdminString
FROM SNMP-FRAMEWORK-MIB;

jcp ::= {enterprises xxx}

-- to be obtained from IANA by JCP PMO

j2eeMIB MODULE-IDENTITY

LAST-UPDATED "200112140000Z"

ORGANIZATION "JSR 77 Expert Group"

CONTACT-INFO

"WG EMail: <http://groups.yahoo.com/group/eemgmt-experts>

See also: <http://www.jcp.org/jsr/detail/77.jsp>

Author: Anthony Daniel

EMail: jsr77work@yahoo.com"

DESCRIPTION

"This MIB module defines a set of objects for monitoring
the J2EE platform."

REVISION "200107020000Z" -- July 2, 2001

DESCRIPTION

"Initial version, for expert group review."

REVISION "200112170000Z" -- December 17, 2001

DESCRIPTION

"Revised version, for public review. Updates include the following:

- Added domains, nodes, JVMs.
- Dropped more detailed tables for resources whose instrumentation was eliminated from the model.
- Clarified that state-manageable objects which are stopped can be restarted.
- Introduced a textual convention for managed object names.
- MIB was made private to JCP.org; it can be re-rooted when published as RFC.
- Section 5 added.
- Introductory sections fleshed out.
- Added Enterprise numbers for Domains, Servers, JVMs, to make vendor checks more authoritative for JSR77 clients.
- Added NOTIFICATION-TYPEs (but left OBJECT list empty).
- Some generic managed object and state manageable attributes left unexpanded.
- Still missing JVM, JDBC, and JMS performance statistics; left inconsistent the fact that sometimes augmentation is used, sometimes the objects are just embedded in the table.
- Still missing conformance and compliance information.

"

::= { jcp xxx } -- OID assigned by JCP PMO

J2eeMOName ::= TEXTUAL-CONVENTION

STATUS current

DESCRIPTION "

A textual identifier used for names of J2EE Managed Objects.

To facilitate internationalization, a J2eeMOName is represented using the ISO/IEC IS 10646-1 character set, encoded as an octet string using the UTF-8 transformation format described in [RFC2279].

A J2eeMOName cannot be null, and cannot contain a colon (':'), comma (','), equal sign ('='), asterisk ('*'), or question mark ('?').

"

SYNTAX OCTET STRING (SIZE (1..255))

J2eeSMState ::= TEXTUAL-CONVENTION

STATUS current

DESCRIPTION "

The operational status of a 'StateManageable' object within the JSR77 model. Six states are possible. The three core states are RUNNING, STOPPED, and FAILED. Extended states are STARTING, STOPPING, and OTHER. Core states MUST be supported; extended states SHOULD be supported.

RUNNING is the normal operational status of a StateManageable object. It indicates that the StateManageable object is fully operational from the perspective of the JSR77 model. Thus requests on MIB objects associated with this StateManageable object will return results as defined by the JSR77 model.

STOPPED indicates that the associated StateManageable object is quiescent, inactive, or nonexistent. It also indicates that the associated object can be restarted. Requests on MIB objects associated with this StateManageable object will return results as defined by the JSR77 model.

FAILED is exactly the same as STOPPED except that the StateManageable object has become stopped for unexpected reasons, and some form of intervention, perhaps out-of-band, is required.

STARTING indicates that the associated StateManageable object has begun the process of becoming fully operational. This state indicates that the StateManageable object is in a setup or initialization phase, and consequently has undefined behavior from the perspective of the JSR77 model. Requests on

MIB objects associated with this StateManageable object may return valid results, invalid results, or errors.

STOPPING indicates that the associated StateManageable object has begun the process of becoming quiescent, inactive, or nonexistent. Requests on MIB objects associated with this StateManageable object may return valid results, invalid results, or errors.

OTHER means none of the above apply.

"

SYNTAX INTEGER { other(1),
failed(2), stopped(3), running(4),
starting(5), stopping(6) }

--

-- The groups defined within this MIB module:

--

j2eeObjects OBJECT IDENTIFIER ::= { j2eeMIB 1 }
j2eeNotifications OBJECT IDENTIFIER ::= { j2eeMIB 2 }
j2eeConformance OBJECT IDENTIFIER ::= { j2eeMIB 3 }

--

--

--

j2eeDomTable OBJECT-TYPE

SYNTAX SEQUENCE OF J2eeDomEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This table lists the J2EE Management Domains which have a presence on a system."

```
::= { j2eeObjects 1 }
```

j2eeDomEntry OBJECT-TYPE

SYNTAX J2eeDomEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"An entry describing a J2EE Management Domain."

INDEX { j2eeDomIndex }

```
::= { j2eeDomTable 1 }
```

J2eeDomEntry ::= SEQUENCE {

j2eeDomIndex Integer32,

j2eeMODomName J2eeMOMName,

j2eeDomEnterprise Integer32,

j2eeMDomStateManaged TruthValue,

j2eeMODomStatProv TruthValue,

j2eeMODomEventProv TruthValue,

j2eeSMDomState J2eeSMState,

j2eeSMDomStartTime J2eeSMStartTime

}

j2eeDomIndex OBJECT-TYPE

SYNTAX Integer32 (1..2147483647)

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"The locally arbitrary, but unique identifier associated with each J2EE management domain entry.

The value is expected to remain constant at least from one re-initialization of the entity's management system to the next re-initialization."

```
::= { j2eeDomEntry 1 }
```

j2eeDomMOMName OBJECT-TYPE

SYNTAX J2eeMOMName

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The name of the domain. This attribute is required because the JSR77 model requires servers to be managed objects, and all managed objects have names."

::= { j2eeDomEntry 2 }

j2eeDomEnterprise OBJECT-TYPE

SYNTAX Integer32 (1..2147483647)

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The SMI Network Management Private Enterprise Code assigned by IANA to the vendor providing the J2EE Management software for this domain."

::= { j2eeDomEntry 3 }

j2eeMODomStateManaged OBJECT-TYPE

SYNTAX TruthValue

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This flag is True if the domain is state-manageable as defined by the JSR77 model for all managed objects."

::= { j2eeDomEntry 4 }

j2eeMODomStatProv OBJECT-TYPE

SYNTAX TruthValue,

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This flag is True if the domain provides statistics as defined by the JSR77 model for managed objects."

::= { j2eeDomEntry 5 }

j2eeMODOMEventProv OBJECT-TYPE

SYNTAX TruthValue,

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This flag is True if the domain supports event notification and is currently configured to send traps for all events defined and available from all event providers in the domain."

::= { j2eeDomEntry 6 }

j2eeSMDomState OBJECT-TYPE

SYNTAX J2eeSMState,

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The status of the domain, which is based on the state of all of the servers within the domain. If all servers are in the running(4) state, the domain's state is running(4). Otherwise, the domain's state is the least functional state that one or more of its servers is at. The states, ordered from most to least functional are: running(4), starting(5), other(1), stopping(6), stopped(3), failed(2)."

::= { j2eeDomEntry 7 }

j2eeSMDomStartTime OBJECT-TYPE

SYNTAX DateAndTime

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A sequence of octets, following the SNMPv2-TC DateAndTime format, representing the date and time, to the nearest hundredth of a second, at which the domain has most recently entered the running(4) state."

::= { j2eeDomEntry 8 }

--

-- ...

--

j2eeSrvTable OBJECT-TYPE

SYNTAX SEQUENCE OF J2eeSrvEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This table lists the J2EE Servers present on a system."

::= { j2eeObjects 2 }

j2eeSrvEntry OBJECT-TYPE

SYNTAX J2eeSrvEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"An entry describing a J2EE Server in a J2EE Management Domain."

INDEX { j2eeDomIndex, j2eeSrvIndex }

::= { j2eeSrvTable 1 }

J2eeSrvEntry ::= SEQUENCE {

j2eeSrvIndex Integer32,

j2eeMOSrvName J2eeMOName,

j2eeSrvEnterprise Integer32,

j2eeSrvVendor SnmpAdminString,

j2eeMOSrvStateManaged TruthValue,

j2eeMOSrvStatProv TruthValue,

j2eeMOSrvEventProv TruthValue,

j2eeSMSrvState J2eeSMState,

j2eeSMSrvStartTime J2eeSMStartTime

}

j2eeSrvIndex OBJECT-TYPE

SYNTAX Integer32 (1..2147483647)

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"The locally arbitrary, but unique identifier associated with each J2EE server entry.

The value is expected to remain constant at least from one re-initialization of the entity's management system to the next re-initialization."

::= { j2eeSrvEntry 1 }

j2eeMOSrvName OBJECT-TYPE

SYNTAX J2eeMOName

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The name of the server. This attribute is required because the JSR77 model requires servers to be managed objects, and all managed objects have names."

::= { j2eeSrvEntry 2 }

j2eeSrvEnterprise OBJECT-TYPE

SYNTAX Integer32 (1..2147483647)

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The SMI Network Management Private Enterprise Code assigned by IANA to the J2EE platform vendor for the server."

::= { j2eeSrvEntry 3 }

j2eeSrvVendor OBJECT-TYPE

SYNTAX SnmpAdminString

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A human-readable string identifying the J2EE platform vendor for the server."

::= { j2eeSrvEntry 5 }

j2eeMOSrvStatProv OBJECT-TYPE

SYNTAX TruthValue,

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This flag is True if the server provides statistics as defined by the JSR77 model for managed objects."

::= { j2eeSrvEntry 6 }

j2eeMOSrvEventProv OBJECT-TYPE

SYNTAX TruthValue,

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"This flag is True if the server provides events as defined by the JSR77 model for managed objects."
 ::= { j2eeSrvEntry 7 }

j2eeSMSrvState OBJECT-TYPE

SYNTAX J2eeSMState,
 MAX-ACCESS read-only
 STATUS current
 DESCRIPTION
 "The status of the server, following state-manageable object semantics."
 ::= { j2eeSrvEntry 8 }

j2eeSMSrvStartTime OBJECT-TYPE

SYNTAX DateAndTime
 MAX-ACCESS read-only
 STATUS current
 DESCRIPTION
 "A sequence of octets, following the SNMPv2-TC DateAndTime format, representing the date and time, to the nearest hundredth of a second, at which the server has most recently entered the running(4) state."
 ::= { j2eeSrvEntry 9 }

--
 -- ...
 --

j2eeNodeTable OBJECT-TYPE

SYNTAX SEQUENCE OF J2eeNodeEntry
 MAX-ACCESS not-accessible
 STATUS current
 DESCRIPTION
 "This table lists all nodes that a particular server spans."
 ::= { j2eeObjects 3 }

j2eeNodeEntry OBJECT-TYPE

SYNTAX J2eeNodeEntry
 MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"An entry describing a node from the perspective of a particular J2EE server. Note that a node hosting multiple servers from one or more vendors would have multiple entries."

INDEX { j2eeDomIndex, j2eeSrvIndex, j2eeNodeIndex }

::= { j2eeSrvTable 1 }

J2eeNodeEntry ::= SEQUENCE {

j2eeNodeIndex Integer32,

j2eeNodeName SnmpAdminString, -- not a J2eeMOName

j2eeNodeHWType SnmpAdminString,

-- j2eeNodeHWProductID OBJECT IDENTIFIER, -- use this or ProductID TC

-- -- to align with Host MIB RFC 2790

j2eeNodeOSType SnmpAdminString,

-- j2eeNodeOSProductID OBJECT IDENTIFIER, -- to align with Host MIB
RFC 2790

j2eeNodeIPAddress TAddress

-- The following are implied by the J2EE management model

-- j2eeMONodeStateManaged TruthValue,

-- j2eeMONodeStatProv TruthValue,

-- j2eeMONodeEventProv TruthValue,

-- j2eeSMNodeState J2eeSMState,

-- j2eeSMNodeStartTime J2eeSMStartTime

}

j2eeNodeIndex OBJECT-TYPE

SYNTAX Integer32 (1..2147483647)

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"The locally arbitrary, but unique identifier associated with each J2EE node entry.

The value is expected to remain constant at least from one re-initialization of the entity's management system to the next re-initialization."

::= { j2eeNodeEntry 1 }

j2eeNodeName OBJECT-TYPE

SYNTAX SnmpAdminString

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A human-readable string identifying a node hosting a particular J2EE server."

::= { j2eeNodeEntry 2 }

j2eeNodeHWType OBJECT-TYPE

SYNTAX SnmpAdminString

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A human-readable string identifying the type of hardware hosting a particular J2EE server."

::= { j2eeNodeEntry 3 }

j2eeNodeOSType OBJECT-TYPE

SYNTAX SnmpAdminString

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A human-readable string identifying the type of operating system hosting a particular J2EE server."

::= { j2eeNodeEntry 4 }

j2eeNodeIPAddress OBJECT-TYPE

SYNTAX TAddress

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A human-readable string identifying one of the IP addresses known by a J2EE server for a node. If a node is known by multiple IP addresses, multiple entries will exist with the same j2eeNodeName."

::= { j2eeNodeEntry 5 }

--

-- ...

--

j2eeJVMTTable OBJECT-TYPE

SYNTAX SEQUENCE OF J2eeJVMTEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This table lists all JVMs instances that a particular server utilizes."

::= { j2eeObjects 4 }

j2eeJVMTEntry OBJECT-TYPE

SYNTAX J2eeJVMTEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"An entry for a JVM utilized by a particular J2EE server."

INDEX { j2eeDomIndex, j2eeSrvIndex, j2eeNodeIndex, j2eeJVMTIndex }

::= { j2eeSrvTable 1 }

J2eeJVMTEntry ::= SEQUENCE {

j2eeJVMTIndex Integer32,

j2eeJVMTVersion SnmpAdminString,

j2eeJVMTVendor SnmpAdminString,

j2eeJVMTEnterprise Integer32

-- The following are implied by the J2EE management model

-- j2eeMOJVMName J2eeMOName, -- JVM don't need no stinkin' name

-- j2eeMOJVMStateManaged TruthValue,

-- j2eeMOJVMStatProv TruthValue,

-- j2eeMOJVMEventProv TruthValue,

-- j2eeSMJVMState J2eeSMState,

-- j2eeSMJVMStartTime J2eeSMStartTime

}

j2eeJVMTIndex OBJECT-TYPE

SYNTAX Integer32 (1..2147483647)

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"The locally arbitrary, but unique identifier associated with each J2EE JVM entry. Note that the existence of this index permits a server to make use of multiple JVMs on a single node.

- (If multi-JVMs-per-node never really happens amongst J2EE
- platform providers, some efficiency would be gained by
- dropping this object and using j2eeNodeIndex directly for
- the j2eeJVMTTable.)

The value is expected to remain constant at least from one re-initialization of the entity's management system to the next re-initialization."

::= { j2eeNodeEntry 1 }

j2eeJVMVersion OBJECT-TYPE

SYNTAX SnmpAdminString

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A human-readable string identifying the Java Runtime Environment version of the JVM. It must be identical to the value of the system property java.version."

::= { j2eeNodeEntry 2 }

j2eeJVMVendor OBJECT-TYPE

SYNTAX SnmpAdminString

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"A human-readable string identifying the Java Runtime Environment vendor of the JVM. It must be identical to the value of the system property java.vendor."

::= { j2eeNodeEntry 3 }

j2eeJVMEnterprise OBJECT-TYPE

SYNTAX Integer32 (1..2147483647)

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The SMI Network Management Private Enterprise Code assigned
by IANA to the Java Runtime Environment vendor of the JVM."

::= { j2eeNodeEntry 5 }

--

-- ...

--

j2eeAppTable OBJECT-TYPE

SYNTAX SEQUENCE OF J2eeAppEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This table lists the J2EE Applications present on a system."

::= { j2eeObjects 5 }

j2eeAppEntry OBJECT-TYPE

SYNTAX J2eeAppEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"An entry describing a J2EE Application in a J2EE Management Domain."

INDEX { j2eeDomIndex, j2eeAppIndex }

::= { j2eeAppTable 1 }

J2eeAppEntry ::= SEQUENCE {

j2eeAppIndex Integer32,

j2eeMOAppName J2eeMOName,

j2eeMOAppStateManaged TruthValue,

j2eeMOAppStatProv TruthValue,

j2eeMOAppEventProv TruthValue,

j2eeSMApState J2eeSMState,

j2eeSMApStartTime J2eeSMStartTime

}

```

j2eeAppIndex OBJECT-TYPE -- expanded verbiage to be filled in later
    ::= { j2eeModEntry 1 }
j2eeMOAppName OBJECT-TYPE -- expanded verbiage to be filled in later
    ::= { j2eeModEntry 2 }
j2eeMOAppStateManaged OBJECT-TYPE -- expanded verbiage to be filled in
later
    ::= { j2eeModEntry 3 }
j2eeMOAppStatProv    OBJECT-TYPE -- expanded verbiage to be filled in later
    ::= { j2eeModEntry 4 }
j2eeMOAppEventProv  OBJECT-TYPE -- expanded verbiage to be filled in later
    ::= { j2eeModEntry 5 }
j2eeSMAppState      OBJECT-TYPE -- expanded verbiage to be filled in later
    ::= { j2eeModEntry 6 }
j2eeSMAppStartTime  OBJECT-TYPE -- expanded verbiage to be filled in later
    ::= { j2eeModEntry 7 }

```

```

--
--
--

```

```

j2eeModTable OBJECT-TYPE
    SYNTAX      SEQUENCE OF J2eeModEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "This table lists the modules in all applications deployed in
        in a domain."
    ::= { j2eeObjects 6 }

```

```

j2eeModEntry OBJECT-TYPE
    SYNTAX      J2eeModEntry
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "An entry for each module in each application in a J2EE Management
        Domain. The entry is indexed by the J2EE application index as well
        as a unique ID per module."

```

```
INDEX { j2eeDomIndex, j2eeAppIndex, j2eeModIndex }
::= { j2eeModTable 1 }
```

```
J2eeModEntry ::= SEQUENCE {
    j2eeModIndex      Integer32,
    j2eeMOModName     J2eeMOMName,
    j2eeModType       INTEGER,
    j2eeModJVMIndex   Integer32,
    j2eeMOModStateManaged TruthValue,
    j2eeMOModStatProv TruthValue,
    j2eeMOModEventProv TruthValue,
    j2eeSMModState     J2eeSMState,
    j2eeSMModStartTime J2eeSMStartTime
}
```

```
j2eeModIndex OBJECT-TYPE
    SYNTAX      Integer32 (1..2147483647)
    MAX-ACCESS  not-accessible
    STATUS      current
    DESCRIPTION
        "The locally arbitrary, but unique identifier associated
        with each J2EE module.
```

The value is expected to remain constant at least from one re-initialization of the entity's management system to the next re-initialization."

```
::= { j2eeModEntry 1 }
```

```
j2eeMOModName OBJECT-TYPE
    SYNTAX      J2eeMOMName
    MAX-ACCESS  read-only
    STATUS      current
    DESCRIPTION
        "The name of the module. This attribute is required because
        the JSR77 model requires modules to be managed objects,
        and all managed objects have names."
    ::= { j2eeModEntry 2 }
```


j2eeModType OBJECT-TYPE

```
SYNTAX    INTEGER {
        other(1), ejb(2), web(3), connector(4)
    }
MAX-ACCESS read-only
STATUS     current
DESCRIPTION
 ::= { j2eeModEntry 3 }
```

j2eeModJVMIndex OBJECT-TYPE

```
SYNTAX    Integer32 (1..2147483647)
MAX-ACCESS read-only
STATUS     current
DESCRIPTION
    "The arbitrary, but unique identifier for the JVM on which
    each J2EE module is running.
```

```
    The value MUST be able to be used to query the j2eeJVMTTable
    for details about the JVM."
 ::= { j2eeModEntry 4 }
```

j2eeMOModStateManaged OBJECT-TYPE -- expanded verbiage to be filled in later

```
 ::= { j2eeModEntry 5 }
```

j2eeMOModStatProv OBJECT-TYPE -- expanded verbiage to be filled in later

```
 ::= { j2eeModEntry 6 }
```

j2eeMOModEventProv OBJECT-TYPE -- expanded verbiage to be filled in later

```
 ::= { j2eeModEntry 7 }
```

j2eeSMModState OBJECT-TYPE -- expanded verbiage to be filled in later

```
 ::= { j2eeModEntry 8 }
```

j2eeSMModStartTime OBJECT-TYPE -- expanded verbiage to be filled in later

```
 ::= { j2eeModEntry 9 }
```

```
--
--
--
```

j2eeBeanTable OBJECT-TYPE

SYNTAX SEQUENCE OF J2eeBeanEntry
 MAX-ACCESS not-accessible
 STATUS current
 DESCRIPTION
 "This table lists the EJBs in all applications deployed in
 a domain."
 ::= { j2eeObjects 7 }

j2eeBeanEntry OBJECT-TYPE
 SYNTAX J2eeBeanEntry
 MAX-ACCESS not-accessible
 STATUS current
 DESCRIPTION
 "An entry for each bean in each application in a J2EE Management
 Domain. The entry is indexed by the J2EE application index, the
 module index, and a unique ID per bean."
 INDEX { j2eeDomIndex, j2eeAppIndex, j2eeModIndex, j2eeBeanIndex }
 ::= { j2eeBeanTable 1 }

J2eeBeanEntry ::= SEQUENCE {
 j2eeBeanIndex Integer32,
 j2eeMOBeanName J2eeMOName,
 j2eeBeanType Integer32,
 j2eeMOBeanStateManaged TruthValue,
 j2eeMOBeanStatProv TruthValue,
 j2eeMOBeanEventProv TruthValue,
 j2eeSMBeanState J2eeSMState,
 j2eeSMBeanStartTime J2eeSMStartTime
 }

j2eeBeanType OBJECT-TYPE
 SYNTAX INTEGER {
 other(1), entity(2), messageDriven(3),
 stateful(4), stateless(5)
 }
 MAX-ACCESS read-only
 STATUS current
 DESCRIPTION
 ::= { j2eeBeanEntry 3 }

```
--
--
--
```

j2eeSletTable OBJECT-TYPE

SYNTAX SEQUENCE OF J2eeSletEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This table lists the servlets in all applications deployed in a domain."

::= { j2eeObjects 8 }

j2eeSletEntry OBJECT-TYPE

SYNTAX J2eeSletEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"An entry for each servlet in each application in a J2EE Management Domain. The entry is indexed by the J2EE application index, the module index, and a unique ID per servlet."

INDEX { j2eeDomIndex, j2eeAppIndex, j2eeModIndex, j2eeSletIndex }

::= { j2eeSletTable 1 }

J2eeSletEntry ::= SEQUENCE {

j2eeSletIndex Integer32,

j2eeMOSletName J2eeMOName,

j2eeMOSletStateManaged TruthValue,

j2eeMOSletStatProv TruthValue,

j2eeSletServiceCount Counter32, -- only maintained if StatProv is true

j2eeSletServiceMaxTime Gauge32, -- only maintained if StatProv is true

j2eeSletServiceMinTime Gauge32, -- only maintained if StatProv is true

j2eeSletServiceTotal Counter32, -- only maintained if StatProv is true

j2eeMOSletEventProv TruthValue,

j2eeSMSletState J2eeSMState,

j2eeSMSletStartTime J2eeSMStartTime

}

```

j2eeSletIndex OBJECT-TYPE -- expanded verbiage to be filled in later
 ::= { j2eeSletEntry 1 }
j2eeMOSletName OBJECT-TYPE -- expanded verbiage to be filled in later
 ::= { j2eeSletEntry 2 }
j2eeMOSletStateManaged OBJECT-TYPE -- expanded verbiage to be filled in
later
 ::= { j2eeSletEntry 3 }
j2eeMOSletStatProv  OBJECT-TYPE -- expanded verbiage to be filled in later
 ::= { j2eeSletEntry 4 }
j2eeSletServiceCount OBJECT-TYPE -- expanded verbiage to be filled in later
 ::= { j2eeSletEntry 5 }
j2eeSletServiceMaxTime OBJECT-TYPE -- expanded verbiage to be filled in
later
 ::= { j2eeSletEntry 6 }
j2eeSletServiceMinTime OBJECT-TYPE -- expanded verbiage to be filled in
later
 ::= { j2eeSletEntry 7 }
j2eeSletServiceTotal  OBJECT-TYPE -- expanded verbiage to be filled in later
 ::= { j2eeSletEntry 8 }
j2eeMOSletEventProv  OBJECT-TYPE -- expanded verbiage to be filled in later
 ::= { j2eeSletEntry 9 }
j2eeSMSletState      OBJECT-TYPE -- expanded verbiage to be filled in later
 ::= { j2eeSletEntry 10 }
j2eeSMSletStartTime  OBJECT-TYPE -- expanded verbiage to be filled in later
 ::= { j2eeSletEntry 11 }

--
--
--

j2eeAdapTable OBJECT-TYPE
  SYNTAX      SEQUENCE OF J2eeAdapEntry
  MAX-ACCESS  not-accessible
  STATUS      current
  DESCRIPTION
    "This table lists the resource adapters in all applications
    deployed in a domain."
  ::= { j2eeObjects 9 }

```

j2eeAdapEntry OBJECT-TYPE

SYNTAX J2eeAdapEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"An entry for each resource adapter in each application in a J2EE Management Domain. The entry is indexed by the J2EE application index, the module index, and a unique ID per adapter."

INDEX { j2eeDomIndex, j2eeAppIndex, j2eeModIndex, j2eeAdapIndex }

::= { j2eeAdapTable 1 }

J2eeAdapEntry ::= SEQUENCE {

j2eeAdapIndex Integer32,

j2eeMOAdapName J2eeMOName,

j2eeMOAdapStateManaged TruthValue,

j2eeMOAdapStatProv TruthValue,

j2eeMOAdapEventProv TruthValue,

j2eeSMAdapState J2eeSMState,

j2eeSMAdapStartTime J2eeSMStartTime

}

j2eeAdapIndex OBJECT-TYPE -- expanded verbiage to be filled in later

::= { j2eeAdapEntry 1 }

j2eeMOAdapName OBJECT-TYPE -- expanded verbiage to be filled in later

::= { j2eeAdapEntry 2 }

j2eeMOAdapStateManaged OBJECT-TYPE -- expanded verbiage to be filled in later

::= { j2eeAdapEntry 3 }

j2eeMOAdapStatProv OBJECT-TYPE -- expanded verbiage to be filled in later

::= { j2eeAdapEntry 4 }

j2eeMOAdapEventProv OBJECT-TYPE -- expanded verbiage to be filled in later

::= { j2eeAdapEntry 5 }

j2eeSMAdapState OBJECT-TYPE -- expanded verbiage to be filled in later

::= { j2eeAdapEntry 6 }

j2eeSMAdapStartTime OBJECT-TYPE -- expanded verbiage to be filled in later

::= { j2eeAdapEntry 7 }

```
--
--
--
```

j2eeRsrcTable OBJECT-TYPE

SYNTAX SEQUENCE OF J2eeRsrcEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This table lists the resources used by a server to provide J2EE standard services."

::= { j2eeObjects 10 }

j2eeRsrcEntry OBJECT-TYPE

SYNTAX J2eeRsrcEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"An entry for each resource used in each server in a J2EE Management Domain. These resources are used by the J2EE core sever to provide the J2EE standard services required by the J2EE platform architecture.

The entry is indexed by the J2EE server and a unique ID per resource."

INDEX { j2eeDomIndex, j2eeSrvIndex, j2eeRsrcIndex }

::= { j2eeRsrcTable 1 }

J2eeRsrcEntry ::= SEQUENCE {

j2eeRsrcIndex Integer32,

j2eeMORsrcName J2eeMOName,

j2eeRsrcType INTEGER,

j2eeMORsrcStateManaged TruthValue,

j2eeMORsrcStatProv TruthValue,

j2eeMORsrcEventProv TruthValue,

j2eeSMRsrcState J2eeSMState,

j2eeSMRsrcStartTime J2eeSMStartTime

}

j2eeRsrcIndex OBJECT-TYPE

SYNTAX Integer32 (1..2147483647)

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"The locally arbitrary, but unique identifier associated with each J2EE resource.

The value is expected to remain constant at least from one re-initialization of the entity's management system to the next re-initialization."

::= { j2eeRsrcEntry 1 }

j2eeMORsrcName OBJECT-TYPE

SYNTAX J2eeMOName

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The name of the resource. This attribute is required because the JSR77 model requires resources to be managed objects, and all managed objects have names."

::= { j2eeRsrcEntry 2 }

j2eeRsrcType OBJECT-TYPE

SYNTAX INTEGER {
 other(1), javamail(2), jdbc(3), jms(4),
 jndi(5), jta(6), rmiiop(7), url(8)
 }

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The type of resource:

 javamail(2) JavaMail(tm)

 jdbc(3) Java Database Connectivity

 jms(4) Java Message Service

 jndi(5) Java Naming and Directory Interface(tm)

 jta(6) Java Transaction API

 rmiiop(7) Remote Method Invocation over Internet
 Inter-ORB Protocol

```

        url(8)      URL"
 ::= { j2eeRsrcEntry 3 }

```

j2eeMORsrcStateManaged OBJECT-TYPE -- expanded verbiage to be filled in later

```
 ::= { j2eeRsrcEntry 4 }

```

j2eeMORsrcStatProv OBJECT-TYPE -- expanded verbiage to be filled in later

```
 ::= { j2eeRsrcEntry 5 }

```

j2eeMORsrcEventProv OBJECT-TYPE -- expanded verbiage to be filled in later

```
 ::= { j2eeRsrcEntry 6 }

```

j2eeSMRsrcState OBJECT-TYPE -- expanded verbiage to be filled in later

```
 ::= { j2eeRsrcEntry 7 }

```

j2eeSMRsrcStartTime OBJECT-TYPE -- expanded verbiage to be filled in later

```
 ::= { j2eeRsrcEntry 8 }

```

```
--
```

```
--
```

```
--
```

j2eeJDBCTable OBJECT-TYPE

SYNTAX SEQUENCE OF J2eeJDBCEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This table lists the JDBC Drivers and Data Sources used by a server to provide J2EE standard services."

```
 ::= { j2eeObjects 11 }

```

j2eeJDBCEntry OBJECT-TYPE

SYNTAX J2eeJDBCEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"An entry for each JDBC Driver and Data Source pair used in each server in a J2EE Management Domain. The entry is indexed by the J2EE server and a unique ID per pair."

```
 INDEX { j2eeDomIndex, j2eeSrvIndex, j2eeJDBCIndex }

```



```
::= { j2eeJDBCTable 1 }
```

```
J2eeJBCDriverEntry ::= SEQUENCE {
    j2eeJBCDriverIndex      Integer32,
    j2eeMOJBCDriverName     J2eeMOName,
    j2eeMOJBCSourceName     J2eeMOName,
    j2eeMOJBCStateManaged  TruthValue,
    j2eeMOJBCStatProv       TruthValue,
    j2eeMOJBCEventProv      TruthValue,
    j2eeSMJBCState          J2eeSMState,
    j2eeSMJBCStartTime      J2eeSMStartTime
}
```

j2eeJBCIndex OBJECT-TYPE

SYNTAX Integer32 (1..2147483647)

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"The locally arbitrary, but unique identifier associated with each J2EE resource.

The value is expected to remain constant at least from one re-initialization of the entity's management system to the next re-initialization."

```
::= { j2eeRsrcEntry 1 }
```

j2eeMOJBCDriverName OBJECT-TYPE

SYNTAX J2eeMOName

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The name of the JDBC Driver. This attribute is required because the JSR77 model requires resources to be managed objects, and all managed objects have names."

```
::= { j2eeJBCEntry 2 }
```

j2eeMOJBCSourceName OBJECT-TYPE

SYNTAX J2eeMOName

MAX-ACCESS read-only

STATUS current

DESCRIPTION

"The name of the JDBC Data Source. This attribute is required because the JSR77 model requires resources to be managed objects, and all managed objects have names."

::= { j2eeJDBCEntry 2 }

j2eeMOJDBCStateManaged OBJECT-TYPE -- expanded verbiage to be filled in later

::= { j2eeJDBCEntry 4 }

j2eeMOJDBCStatProv OBJECT-TYPE -- expanded verbiage to be filled in later

::= { j2eeJDBCEntry 5 }

j2eeMOJDBCEventProv OBJECT-TYPE -- expanded verbiage to be filled in later

::= { j2eeJDBCEntry 6 }

j2eeSMJDBCState OBJECT-TYPE -- expanded verbiage to be filled in later

::= { j2eeJDBCEntry 7 }

j2eeSMJDBCStartTime OBJECT-TYPE -- expanded verbiage to be filled in later

::= { j2eeJDBCEntry 8 }

--

--

--

--

-- The next few tables are the performance-augmented versions
-- of previous tables. These are only supported when the
-- corresponding managed object is a statistics provider.

--

j2eePDFBeanTable OBJECT-TYPE

SYNTAX SEQUENCE OF J2eePDFBeanTableEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This table augments the J2EE bean table with performance data framework statistics objects. Some statistics will have different interpretations depending on the type of bean being augmented."

::= { j2eeObjects 12 }

j2eePDFBeanTableEntry OBJECT-TYPE

SYNTAX J2eePDFBeanTableEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"The performance-augmented version of the J2EE Bean table. This entry extends each row in the Bean table with EJB-specific performance statistics."

AUGMENT {j2eeBeanTableEntry}

::= { j2eePDFBeanTable 1 }

J2eePDFBeanTableEntry ::= SEQUENCE {

j2eePDFCreateCount Counter32, -- all bean types

j2eePDFRemoveCount Counter32, -- all bean types

j2eePDFReadyCount Gauge32, -- entity or session only

j2eePDFPooledOrPassCount Gauge32, -- entity or stateful session only

j2eePDFMessageCount Counter32 -- message-driven only

}

j2eePDFCreateCount OBJECT-TYPE -- expanded verbiage to be filled in later

::= { J2eePDFBeanTableEntry 1 }

j2eePDFRemoveCount OBJECT-TYPE -- expanded verbiage to be filled in later

::= { J2eePDFBeanTableEntry 2 }

j2eePDFReadyCount OBJECT-TYPE -- expanded verbiage to be filled in later

::= { J2eePDFBeanTableEntry 3 }

j2eePDFPooledOrPassCount OBJECT-TYPE -- expanded verbiage to be filled in later

::= { J2eePDFBeanTableEntry 4 }

j2eePDFMessageCount OBJECT-TYPE -- expanded verbiage to be filled in later

::= { J2eePDFBeanTableEntry 5 }

--
--
--

j2eePDFJavaMailTable OBJECT-TYPE

SYNTAX SEQUENCE OF J2eePDFJavaMailTableEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This table augments the Resource table with performance data framework statistics objects for JavaMail resources."

::= { j2eeObjects 13 }

j2eePDFJavaMailTableEntry OBJECT-TYPE

SYNTAX J2eePDFJavaMailTableEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"The performance-augmented, JavaMail-specific version of the J2EE Resource table. This entry extends each row in the resource table with JavaMail-specific performance statistics. Note that this augmentation only applies to rows in the J2EE resource table where j2eeRsrcType is javamail(2); attempts to access the corresponding instance of any of these objects when j2eeModType is not javamail(2) may result in either noSuchName (SNMPv1) or noSuchInstance (SNMPv2) being returned by the agent."

AUGMENT {j2eeRsrcTableEntry}

::= { j2eePDFJavaMailTable 1 }

J2eePDFJavaMailTableEntry ::= SEQUENCE {

j2eePDFSentCount Counter32
}

j2eePDFSentCount OBJECT-TYPE -- expanded verbiage to be filled in later

::= { J2eePDFJavaMailTableEntry 1 }

--
--
--

j2eePDFJTATable OBJECT-TYPE

SYNTAX SEQUENCE OF J2eePDFJTATableEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"This table augments the Resource table with performance data framework statistics objects for JTA resources."

::= { j2eeObjects 14 }

j2eePDFJTATableEntry OBJECT-TYPE

SYNTAX J2eePDFJTATableEntry

MAX-ACCESS not-accessible

STATUS current

DESCRIPTION

"The performance-augmented, JTA-specific version of the J2EE Resource table. This entry extends each row in the resource table with JTA-specific performance statistics. Note that this augmentation only applies to rows in the J2EE resource table where j2eeRsrcType is jta(6); attempts to access the corresponding instance of any of these objects when j2eeModType is not jta(6) may result in either noSuchName (SNMPv1) or noSuchInstance (SNMPv2) being returned by the agent."

AUGMENT {j2eeRsrcTableEntry}

::= { j2eePDFJTATable 1 }

J2eePAJavaMailTableEntry ::= SEQUENCE {

j2eePDFActiveCount Gauge32,

j2eePDFCommittedCount Counter32,

j2eePDFRolledBackCount Counter32

}

j2eePDFActiveCount OBJECT-TYPE -- expanded verbiage to be filled in later

::= { J2eePDFActiveTableEntry 1 }

j2eePDFCommittedCount OBJECT-TYPE -- expanded verbiage to be filled in later

::= { J2eePDFActiveTableEntry 2 }

```
j2eePDFRolledBackCount OBJECT-TYPE -- expanded verbiage to be filled in
later
 ::= { J2eePDFActiveTableEntry 3 }

--
-- Notifications
--

j2eeServerStateChange NOTIFICATION-TYPE
    OBJECTS { }
    STATUS current
    DESCRIPTION "A server entered RUNNING, STOPPED, or FAILED state."
 ::= { j2eeNotifications 1 }

j2eeEntityBeanStateChange NOTIFICATION-TYPE
    OBJECTS { }
    STATUS current
    DESCRIPTION "An entity bean entered RUNNING, STOPPED, or FAILED
state."
 ::= { j2eeNotifications 2 }

j2eeMessageDrivenBeanStateChange NOTIFICATION-TYPE
    OBJECTS { }
    STATUS current
    DESCRIPTION "A message-driven bean entered RUNNING, STOPPED, or
FAILED state."
 ::= { j2eeNotifications 3 }

j2eeResourceStateChange NOTIFICATION-TYPE
    OBJECTS { }
    STATUS current
    DESCRIPTION "A resource entered RUNNING, STOPPED, or FAILED
state."
 ::= { j2eeNotifications 4 }

j2eeJDBCStateChange NOTIFICATION-TYPE
    OBJECTS { }
    STATUS current
```

DESCRIPTION "A JDBC driver or data source entered RUNNING, STOPPED, or FAILED state."

::= { j2eeNotifications 5 }

-- conformance information

j2eeCompliances OBJECT IDENTIFIER ::= { j2eeConformance 1 }

j2eeGroups OBJECT IDENTIFIER ::= { j2eeConformance 2 }

-- compliance statements

j2eeCompliance MODULE-COMPLIANCE

STATUS current

DESCRIPTION

"The compliance statement for SNMP entities which implement the J2EE MIB."

MODULE -- this module

MANDATORY-GROUPS {
}

-- FIX make this real (i.e. stats provider optional, etc.)

END

JSR77.8.7 IANA Considerations

The Internet Assigned Numbers Authority (IANA) is responsible for maintaining a MIB module which provides OID registrations for well-known languages. The IANA language registry is intended to reduce interoperability problems by providing a single list of well-known languages. However, it is of course still possible to register languages in private OID spaces. Registering languages in private spaces is especially attractive if a language is used for experimentation or if a language is only used in environments where the distribution of MIB modules with the language registration does not cause any maintenance problems.

Any additions or changes to the list of languages registered via IANA require Designated Expert Review as defined in the IANA guidelines [RFC2434]. The Designated Expert will be selected by the IESG Area Director for the IETF Operations and Management Area.

JSR77.8.8 Security Considerations

SNMPv1 by itself is not a secure environment. Even if the network itself is secure (for example by using IPSec), there is no control as to who on the secure network is allowed to access and GET (read) the objects in this MIB.

It is recommended that the implementers consider the security features as provided by the SNMPv3 framework. Specifically, the use of the User-based Security Model RFC 2574 [RFC2574] and the View-based Access Control Model RFC 2575 [RFC2575] is recommended.

It is a customer/user responsibility to ensure that the SNMP entity giving access to an instance of this MIB, is properly configured to give access to the objects only to those principals (users) that have legitimate rights to indeed GET them.

There are no management objects defined in this MIB that have a MAX-ACCESS clause of read-write and/or read-create. If this MIB is implemented correctly, then there is no risk that an intruder can alter or create any management objects of this MIB via direct SNMP SET operations.

It is possible that some information, particularly in the `j2eeRsrcTable`, might be considered sensitive. It is thus important to control even GET access to these objects and possibly to even encrypt the values of these object when sending them over the network via SNMP. Not all implementations of SNMP provide features for such a secure environment.

JSR77.8.9 Intellectual Property

The IETF takes no position regarding the validity or scope of any intellectual property or other rights that might be claimed to pertain to the implementation or use of the technology described in this chapter or the extent to which any license under such rights might or might not be available; neither does it represent that it has made any effort to identify any such rights. Information on the IETF's procedures with respect to rights in standards-track and standards-related documentation can be found in BCP-11. Copies of claims of rights made available for publication and any assurances of licenses to be made available, or the result of an attempt made to obtain a general license or permission for the use of such proprietary rights by implementors or users of this specification can be obtained from the IETF Secretariat.

The IETF invites any interested party to bring to its attention any copyrights, patents or patent applications, or other proprietary rights which may cover technology that may be required to practice this standard. Please address the information to the IETF Executive Director.

JSR77.8.10 Acknowledgments

The J2EE management model this MIB is based on was developed under the Java Community Process (version 2.0) as JSR-077 and includes contributions from the members of the JSR-077 Expert Group (including Hans Hrasna, the spec lead), Sun (as represented by Hans), and other companies. This MIB was written while the author was a member of the JSR-077 Expert Group.

See <http://www.jcp.org/jsr/detail/77.jsp> for further information.

JSR77.8.11 References

All RFCs can be found at <http://www.ietf.org/rfc/rfcNNNN.txt>, replacing NNNN with the appropriate number.

CHAPTER JSR77.9

J2EE Management CIM

This chapter specifies the CIM mapping of the J2EE Management Model. It is optional for J2EE platform implementations to provide a CIM/WBEM implementation of this specification. Implementations that support CIM/WBEM must comply with the CIM model specified in this chapter.

JSR77.9.1 Introduction

The following chapter defines the CIM model for J2EE management. This chapter lays out the CIM model for J2EE Managed Objects (applications, modules, servers etc), events, state management and statistics.

The Managed Object Format (MOF) representation of the model is presented in the “Appendix” on page 136.

JSR77.9.2 Managed Objects

This section deals with modelling basic managed objects such as J2EE servers, applications, modules, resources.

JSR77.9.2.1 J2EEDomain

A domain identifies the scope of the J2EE managed object. As defined in the specification, an instance of J2EEDomain type must be equivalent to the name of the domain it manages. A CIM J2EE_Domain is a subclass of CIM_AdminDomain, which inherits from CIM_System.

JSR77.9.2.2 J2EEServer

The J2EE_Server is a complex system of applications, modules, resources etc. It can be modelled as a subclass of CIM_System called J2EE_Server.

A J2EE server is associated with multiple Java Virtual Machines, nodes, applications, modules and resources.

The following sub-sections will layout the model component by component. The last part of this section talks about the associations between the various parts and the J2EEServer.

JSR77.9.2.2.1 JVM

At a rudimentary level, a JVM is a process which has certain extra information that pertain to Java. Thus, a JVM can be modelled as a subclass of CIM_Setting that is associated with the CIM_Process class through a subclass of the CIM_ElementSetting association. In other words CIM_Process 'has a' J2EE_JVMSetting. J2EE_JVMSetting has properties such as classpath, VM version, etc. So any CIM_Process with a J2EE_JVMSetting associated to it (through a subclass of the CIM_ElementSetting association) is a java process. Note that we do not have to model entities like VM threads, because the CIM_Process class is already associated to CIM_Thread through the CIM_ProcessThread association. In the future, we may want to have a J2EE_JVMThreadSetting per thread of the process to indicate some JVM specific characteristic for the thread.

JSR77.9.2.2.2 Node

A node is a machine on which there are J2EE Server components running. A node is represented as a subclass of CIM_ComputerSystem called J2EE_Node, and is associated through an aggregation, J2EEServerOnNodes, with the J2EE_Server class to represent the multiple nodes that a single J2EE_Server instance could be running on.

JSR77.9.2.2.3 J2EEApplication

A J2EE application is a software entity that is installed on a J2EEServer. An application has two aspects to it - one is the installation/deployment aspect (static) and the second is the runtime aspect.

The installed/deployment aspect is not addressed in this JSR 077, but will be included as part of JSR 088, J2EE Application Deployment.

The runtime/deployed aspect of a J2EE application is modeled in CIM as a J2EE_Application, which is a subclass of the abstract J2EE_DeployedObject class. The J2EE_DeployedObject class is a subclass of CIM_Service and is provided to allow for both J2EE_Application objects as well as J2EE_Module objects. An application can be decomposed into the underlying modules that comprise the application. The modules themselves are subclasses of J2EE_DeployedObject because they can be deployed independent of an J2EE application and associated with multiple J2EE applications. A J2EE_Application is comprised of a number of J2EE_Module(s) and this fact is captured by the J2EE_ServiceComponet association, which is a subclass of CIM_ServiceComponet association. An example of a module would be the EJBModule which could be represented as the leaf level subclass of J2EE_Module.

JSR77.9.2.2.4 J2EEResource

A resource is utilized by a server to provide standard services. A J2EEResource is represented by class J2EE_Resource which is a subclass of CIM_Service. This is an abstract class which has concrete subclasses J2EE_JMSService, J2EE_JNDIService, J2EE_JDBCService etc.

These resource classes are associated with a J2EE_Server through an aggregated association, J2EE_ServerResource.

JSR77.9.3 Events

J2EE.9.3.1 Overview of CIM/WBEM Events

The CIM meta model allows special 'Indication' types to represent events that may be delivered to subscribers. All indications inherit from the CIM indication class, CIM_Indication. The CIM Indications are split into Life Cycle Indications and Process Indications.

'Life Cycle Indications' which are created when CIM classes are created, modified and deleted as well as when instances of CIM classes are created, modified, deleted, read and have methods invoked on them. Process Indications are used to represent indications that cannot be expressed as Life Cycle Indications. The CIM indication class that represents these indications is referred

to as `CIM_ProcessIndication`. The DMTF has defined SNMP traps as sub classes of `CIM_ProcessIndication`.

The subscription process in WBEM consists of associating an instance of `CIM_IndicationFilter` (which specifies the indications of interest) with a `CIM_IndicationHandler` (which represents the destination). A subscriber may create instances of a filter and handler or may choose to instances which already exists. In CIM associating two or more instances is performed by creating an instance of the appropriate association class. In this case, an instance of a `CIM_IndicationSubscription` association is created to associate the filter and destination.

More details of the CIM Event model can be found at <http://www.dmtf.org/education/whitepapers.php>

JSR77.9.3.2 Modeling J2EE Events in CIM

In order to model J2EE Events, we introduce a subclass of `CIM_ProcessIndication` called `J2EE_ManagementIndication`. This subclass has the properties defined in J2EE Management Event with the same semantics. The MOF definition of the class and its superclasses are provided in the “Appendix” on page 136.

Note – Since the `J2EEManagedObject` classes are modeled as CIM classes, it is highly recommended that J2EE instrumentation which maps these classes into CIM should take care of Life Cycle indications on these classes.

JSR77.9.3.3 Handling subscriptions

As described above, a subscription is activated when the `CIM_IndicationFilter` instance is associated to the `CIM_IndicationHandler` instance. The Query string property within `CIM_IndicationFilter` specifies what particular indications are of interest. A query in the WQL query language would typically consist of boolean conditions on the properties of the `J2EE_Indication` class as well as the properties of the referenced `J2EEManagedObject` (which is stored in the `SourceInstance` property as per the MOF). Following is a concrete example:

```
select SourceInstance from J2EE_Indication where J2EE_Indication.Type = 'STATE' and
J2EEIndication.SourceInstance.Name = 'EJB125'
```

The above filter requests the source instance from J2EEEvents whose type is STATE (i.e. these are state manageable events) from the J2EEManagedObject named EJB125. An implementation mapping CIM to J2EE events may add an Event Listener to the appropriate managed object. Once the J2EEEvent is received by the listener can be mapped to a J2EE_Indication and delivered to the destination specified by the CIM_IndicationHandler instance.

JSR77.9.4 State Management

As described in previous sections, every J2EEManagedObject is mapped into a corresponding CIM_LogicalElement. For all those J2EEManagedObject instances which choose to implement the StateManageable interface, the corresponding CIM_LogicalElement instance will be associated to a J2EE_StateManageable instance. The existence of this association signifies to the management application that the given CIM_LogicalElement represents a StateManageable object.

In CIM, instances are associated to each other by creating an instance of the appropriate association class. In this case we introduce a new J2EE_StateManageableLogicalElement association class.

When a management application wishes to perform the StateManageable operations or access the StateManageable properties, it first locates the

CIM_LogicalElement instance representing the J2EEManagedObject and then finds its associated J2EE_StateManageable instance. It may then invoke the appropriate methods and/or access the properties. Details of the J2EE_StateManageable object and J2EE_StateManageableLogicalElement association are provided in the “Appendix” on page 136.

JSR77.9.5 Appendix

JSR77.9.5.1 J2EE Core Model

JSR77.9.5.1.1 J2EE_Domain

```
//=====
// J2EE_Domain
//=====
[Description(
    "Represents the J2EE Domain which comprises"
    "the J2EE Server to be managed as described in the Java 2 "
    "Enterprise Edition Platform specification.")]
class J2EE_Domain : CIM_AdminDomain
{
    [Description(
        "The name of the server domain.")]
    string Domain;
};
```

JSR77.9.5.1.2 J2EE_Server

```
//=====
// J2EE_Server
//=====
[Description(
    "Represents the J2EE Server core of one instance"
    " of a J2EE platform product as described in the Java 2 "
    "Enterprise Edition Platform specification.")]
class J2EE_Server : CIM_System
{
    [Description(
        "The name of the server's vendor.")]
    string Vendor;
};
```


JSR77.9.5.1.3 J2EE_ServersInDomain

```
// =====
// J2EE_ServersInDomain      (Association)
// =====

[Association, Aggregation, Description (
    " The J2EE_ServersInDomain association identifies the J2EE Servers  "
    "that are part of a J2EE Domain for management purposes ") ]
class J2EE_ServersInDomain : CIM_SystemComponent
{
    [Min (1) , Max (1) , Aggregate, Override("Domain")]
    J2EE_Domain REF Domain;
    [Min (0) , Max (NULL), Weak, Override("Server")]
    J2EE_Server REF Server;
};
```

JSR77.9.5.1.4 J2EE_JVMSetting

```
// =====
// J2EE_JVMSetting
// =====

[Description(
    "The settings associated with a"
    " CIM_Process that is the manifestation of a running JVM. The "
    "existence of a setting instance associated to a process instance"
    " indicates that this process is a JVM.") ]
class J2EE_JVMSetting : CIM_Setting
{
    [Key, Description (
        "Name of the Setting instance. This should be the same as the "
        "process id of the JVM process.") ]
    string Name;
    [Key, MaxLen (256), Description (
        "CreationClassName indicates the name of the class or the "
        "subclass used in the creation of an instance. When used "
        "with the other key properties of this class, this property "
        "allows all instances of this class and its subclasses to "
        "be uniquely identified.") ]
    string CreationClassName;
    [Description(
        "The classpath associated with the JVM instance")]
    string Classpath;
    [Description( "The version of the JVM")]
    string Version;
};
```

JSR77.9.5.1.5 J2EE_JVM

```
//=====
// J2EE_JVM    (Association)
//=====
[Association, Description("Represents the association between a
    "running process and an instance of the J2EE_JVMSetting class.) ]
class J2EE_JVMProcessSetting : CIM_ElementSetting
{
    [Key, Override("Element"), Description ("The JVM Process.") ]
    CIM_Process REF Process;
    [Key, Override("Setting"),
        Description ("The Setting object associated with the JVM "
            "process.")]
    J2EE_JVMSetting REF Setting;
}
```

JSR77.9.5.1.6 J2EE_Node

```
//=====
// J2EE_Node
//=====
[Description("The machine on which a server has components running")]
class J2EE_Node : CIM_ComputerSystem
{
};
```

JSR77.9.5.1.7 J2EE_OSONode

```
// =====
// J2EE_OSONode    (Association)
// =====
[Association, Description (
    " The J2EE_OSONode association identifies the OS  that"
    "a specific J2EE Node runs on") ]
class J2EE_ModileInJVM : CIM_InstalledOS
{
    [Min (1) , Max (1) , Override("OS")]
    CIM_OS REF OS;
    [Min (1) , Max (1), Override("Node")]
    J2EE_Node REF Node;
};
```

JSR77.9.5.1.8 J2EE_ServerOnNodes

```
// =====
//  J2EE_ServerOnNodes      (Association)
// =====

[Association, Aggregation, Description (
    “ The J2EE_ServerOnNodes association identifies the Nodes that“
    “a specific J2EE Server runs on, which may be an aggregation since”
    “a single J2EE Server can be distributed on multiple nodes.”) ]
class J2EE_ServerOnNodes : CIM_SystemComponent
{
    [Min (1) , Max (1) , Override(“Server”)]
    J2EE_Server REF Server;
    [Min (0) , Max (NULL), Override(“Node”)]
    J2EE_Node REF Node;
};
```

JSR77.9.5.1.9 J2EE_DeployedObject

```
// =====
//  J2EE_DeployedObject
// =====

[Description (
    “ The J2EE_DeployedObject class is an abstract “

        “class that is used to define J2EE_Module and J2EE_Application “
        “objects that are deployed in the J2EE_Server “) ]
class J2EE_DeployedObject : CIM_Service
{
};
```

JSR77.9.5.1.10 J2EE_Module

```
//=====
//  J2EE_Module
//=====

[Description (“The J2EE_Module class defines a particular function or “
    “capability of an application or application system.”
    “It is expected that specific modules - e.g. EJB, will define”
    “ subclasses of this class”)]
class J2EE_Module : J2EE_DeployedObject
{
};
```

JSR77.9.5.1.11 J2EE_Application

```
//=====
// J2EE_Application
//=====
[Description("The runtime aspects of a J2EE application."
  " The application is decomposed into its functional units using "
    "the J2EE_Module class.")]
class J2EE_Application : J2EE_DeployedObject
{
};
```

JSR77.9.5.1.12 J2EE_ServiceComponent

```
// =====
// J2EE_ServiceComponent      (Association)
// =====
[Association, Aggregation, Description (
  " The J2EE_ServiceComponent association identifies the software "
    "modules for a particular J2EE Application ") ]
class J2EE_ServiceComponent : CIM_ServiceComponent
{
  [Min (1) , Max (1) , Aggregate, Override("Application")]
  J2EE_Application REF Application;
  [Min (0) , Max (NULL), Weak, Override("Component")]
  J2EE_Module REF Module;
};
```

JSR77.9.5.1.13 J2EE_ModuleInJVM

```
// =====
// J2EE_ModuleInJVM          (Association)
// =====
[Association, Aggregation, Description (
  " The J2EE_ModuleInJVM association identifies the JVM that"
    "a specific J2EE Module runs in") ]
class J2EE_ModuleInJVM : CIM_ServiceProcess
{
  [Min (1) , Max (1) , Override("JVM")]
  CIM_Process REF Process;
  [Min (1) , Max (1), Override("Module")]
  J2EE_Module REF Module;
};
```

JSR77.9.5.1.14 J2EE_EJBModule

```
// =====
//  J2EE_EJBModule
// =====
[Description (
    "J2EE_EJBModule is an abstract container class for J2EE_EJBs.")]
class J2EE_EJBModule : J2EE_Module
{
};
```

JSR77.9.5.1.15 J2EE_EJB

```
// =====
//  J2EE_EJB
// =====
[Description (
    "A J2EE_EJB is a component of a J2EE_EJBModule."
    "There can be many EJBs within a Deployed EJB module.")]
class J2EE_EJB : CIM_SoftwareElement
{
};
```

JSR77.9.5.1.16 J2EE_EJBsInModule

```
// =====
//  J2EE_EJBsInModule          (Association)
// =====
[Association, Aggregation, Description (
    "J2EE_EJBsInModule is a aggregation of the EJB Componets"
    "within a Deployed EJB module.")]
class J2EE_EJBsInModule : CIM_SoftwareElementServiceImplementation
{
    [Min (1), Max (NULL), Overrive(" EJB")]
    J2EE_EJB REF EJB
    [Min (1), Max (1), Override("SoftwareElement")]
    CIM_SoftwareElement REF SoftwareElement
};
```

JSR77.9.5.1.17 J2EE_Resource

```
// =====
//  J2EE_Resource
// =====
[Description(
    "Resources associated with a J2EE Server available as"
    " services to other entities. It is expected that specific "
    "J2EE resources like JMS, JDBC etc will create subclasses of"
    "the J2EE_Resource class and add properties relevant to the "
    "specific service.") ]
class J2EE_Resource : CIM_Service
{
};
```

JSR77.9.5.1.18 J2EE_ServerResource

```
// =====
//  J2EE_ServerResource  (Association)
// =====
[Association, Aggregation, Description (
    "J2EE_ServerResource is a subclass of the CIM_ProvidesServiceToElement "
    "association that establishes the relationships between  a J2EE Server"
    "and it's Resources " ]
class J2EE_ServerResource : CIM_ProvidesServiceToElement
{
    [Override ("GroupComponent"), Aggregate, Description (
        "The Server exposing the resources") ]
    J2EE_Server REF Server;
    [Override ("PartComponent"), Description (
        "The resource that is a component of the server.") ]
    J2EE_Resource REF Resource;
};
```

JSR77.9.5.1.19 J2EE_JavaMail

```
//=====
//  J2EE_JavaMail
//=====
[Description (
    "J2EE_JavaMail identifies a JavaMail resource. For each JavaMail resource"
    "available to a server there must be one object that implements the J2EE_JavaMail"
    " model.")]
```

```
class J2EE_JavaMail : J2EE_Resource
{
};
```

JSR77.9.5.1.20 J2EE_JMS

```
//=====
//  J2EE_JMS
//=====
[Description (
    "J2EE_JMS identifies a JMS resource. For each JMS resource provided on a server"
    "there must be one managed object that implements the J2EE_JMS model.")]
```

```
class J2EE_JMS : J2EE_Resource
{
};
```

JSR77.9.5.1.21 J2EE_JNDI

```
//=====
//  J2EE_JNDI
//=====
[Description (
    "J2EE_JNDI identifies a Java Naming and Directory "Interface (JNDI) resource."
    "For each JNDI resource available to a server there must be one managed object"
    " that implements the J2EE_JNDI model.")]
```

```
class J2EE_JNDI : J2EE_Resource
{
};
```

JSR77.9.5.1.22 J2EE_JTA

```
//=====
//  J2EE_JTA
//=====
[Description (
    "J2EE_JTA identifies a Java Transaction API (JTA) resource. For each JTA resource"
    "available to a server there must be one object that implements the J2EE_JTA model.")]
class J2EE_JTA : J2EE_Resource
{
};
```

JSR77.9.5.1.23 J2EE_RMI-IIOP

```
//=====
//  J2EE_RMI-IIOP
//=====
[Description (
    "J2EE_RMI-IIOP identifies an RMI-IIOP resource. For each RMI-IIOP resource"
    "available to a server there must be a managed object that implements the"
    "J2EE_RMI-IIOP model.")]
class J2EE_RMI-IIOP : J2EE_Resource
{
};
```

JSR77.9.5.1.24 J2EE_URL

```
//=====
//  J2EE_URL
//=====
[Description (
    "J2EE_URL identifies a URL resource. For each URL resource available to a server"
    "there must be one managed object that implements the J2EE_URL model.")]
class J2EE_URL : J2EE_Resource
{
};
```


JSR77.9.5.1.25 J2EE_JDBC

```
//=====
//  J2EE_JDBC
//=====
[Description (
    “J2EE_JDBC identifies a JDBC resource. For each JDBC resource provided on a”
    “server there must be one J2EE_JDBC managed object accessible from the
    “servers resources list that identifies it.”)]
class J2EE_JDBC : J2EE_Resource
{
//Properties
string JDBCDataSources[];
};
```

JSR77.9.5.1.26 J2EE_JDBCDataSource

```
//=====
//  J2EE_JDBCDataSource
//=====
[Description (
    “J2EE_JDBCDataSource identifies an individual JDBC driver. For each JDBC”
    “driver being utilized by a server there must be one managed object that implements”
    “the J2EE_JDBCDataSource model.”)]
class J2EE_JDBCDataSource : CIM_LogicalElement
{
};
```

JSR77.9.5.1.27 J2EE_JDBCUsesDataSources

```
//=====
//  J2EE_JDBCUsesDataSources          (Association)
//=====
[Association, Aggregation, Description (
    “J2EE_JDBCUsesDataSources is an association that defines an aggregation of “
    “the JDBC driver resources available to a J2EE_JDBC connector resource.”)]
class J2EE_JDBCUsesDataResources : CIM_Dependency
{
};
```

JSR77.9.5.2 J2EE Events and Indications Model

JSR77.9.5.2.1 *CIM_Indication*

```
//=====
// CIM_Indication          (from the DMTF standard CIMEvent schema)
//=====

[Indication, Abstract, Description (
    "CIM_Indication is the abstract root class for all notifications "
    "about changes in schema, objects and their data, and about "
    "events detected by providers and instrumentation. Subclasses "
    "represent specific types of notifications. \n"
    "\n"
    "To receive an Indication, a consumer (or subscriber) must create "
    "an instance of CIM_IndicationFilter describing the criteria of "
    "the notification, an instance of CIM_IndicationHandler describing "
    "the delivery of the notification, and an instance of "
    "CIM_IndicationSubscription associating the Filter and Handler.") ]
class CIM_Indication
{
//Properties

[Description (
    "The time and date of creation of the Indication. The property "
    "may be set to NULL if the entity creating the Indication is "
    "not capable of determining this information. Note that "
    "IndicationTime may be the same for two Indications that are "
    "generated in rapid succession.") ]
    datetime IndicationTime;
};
```

JSR77.9.5.2.2 *CIM_ProcessIndication*

```
//=====
// CIM_ProcessIndication    (from the DMTF standard CIM Event schema)
//=====

[Indication, Abstract, Description (
    "An abstract superclass for specialized Indication classes, "
    "addressing specific changes and alerts published by providers "
    "and instrumentation. Subclasses include AlertIndication (with "
    "properties such as PerceivedSeverity and ProbableCause), and "
    "SNMPTrapIndication (which recasts Traps as CIM indications).") ]
class CIM_ProcessIndication : CIM_Indication
{
};
```

JSR77.9.5.2.3 J2EE_ManagementIndication

```
//=====
// J2EE_ManagementIndication
//=====

[Indication, Description(
    "Class representing the J2EEEvent class as specified in JSR77. J2EE "
    "managed objects generate events by creating instances of the J2EEEvent "
    "class. Instances of J2EE_Indication class represent the mapping from "
    "J2EE events to the CIM model")])
class J2EE_ManagementIndication : CIM_ProcessIndication
{
    [Description (
        "The time and date of creation of the Indication. The property "
        "may be set to NULL if the entity creating the Indication is "
        "not capable of determining this information. Note that "
        "IndicationTime may be the same for two Indications that are "
        "generated in rapid succession.") ]
    [EmbeddedObject, Description (
        "A copy of the J2EEManagedObject instance that generated this "
        "Indication. For J2EE events only key properties, i.e. name, will "
        "be available for filtering purposes.")])
    string SourceInstance;
    [Description (
        "The sequence number of the indication. Identifies the position of the "
        "indication in a stream of indications. The sequence number provides a "

        "means of determining the order of sequential indications that occurred "
        "with the same timestamp (within the minimum supported unit of time.) ]
    sint64 Sequence;
    [Description (
        "The type of the indication. State manageable managed objects generate "
        "a J2EE_Indication with the type attribute set to "STATE" whenever they "
        "reach the RUNNING, STOPPED or FAILED states") ]
    string Type;
    [Description (
        "An informational message about the indication") ]
    string Message;
};
```

JSR77.9.5.3 J2EE State Model

JSR77.9.5.3.1 J2EE_StateManageable

```
// =====
//  J2EE_StateManageable
// =====

[Description (
    "A J2EE_StateManageable is a Logical Element that represents "
    "the StateManageable features of its associated J2EEManagedObject "
    "An instance of this class must always be associated to a "
    "J2EEManagedObject. This is done through the "
    "J2EE_StateManageableLogicalElement association.") ]
class J2EE_StateManageable : CIM_LogicalElement {

    [Key, Description (
        "Name of the StateManageable instance. This would the same as the name "
        "of the corresponding StateManageable J2EEManagedObject." ) ]
    string Name;
    [Key, MaxLen (256), Description (
        "CreationClassName indicates the name of the class or the "
        "subclass used in the creation of an instance. When used "
        "with the other key properties of this class, this property "
        "allows all instances of this class and its subclasses to "
        "be uniquely identified.") ]
    string CreationClassName;
    [Description (
        "The current state of the associated J2EEManagedObject.")
        Values { "Starting", "Running", "Stopping", "Stopped", "Failed" } ]
    sint32 State;
    [Description (
        "The time that the associated J2EEManagedObject was started.") ]
    datetime StartTime;

    [Description (
        "Starts the associated J2EEManagedObject. This operation can only "
        "be invoked in when State is Stopped. This method, causes State "
        "to become Starting initially, and eventually becomes the RUNNING state. "
        "Note that Start() will not be called on any of the child "
        "StateManageable instances that are registered with this instance. It "
        "is the responsibility of the calling application to start the child "
        "if required. The method returns 0 on success and non-zero for "
        "failure." ) ]
```

```

uint32 Start();
[Description (
    "Starts the associated J2EEManagedObject. This operation can only "
    "be invoked in when State is Stopped. This method, causes State "
    "to become Starting initially, and eventually becomes the RUNNING state. "
    "Additionally, StartRecursive() is called on all the child
    "StateManageable instances that are registered with this instance and "
    "are in the Stopped state. The method returns 0 on success and "
    "non-zero for failure.") ]
uint32 StartRecursive();
[Description (
    "Stops the associated J2EEManagedObject. This operation can only "
    "be invoked when the object is in the RUNNING or STARTING state. This "
    "causes Stop() to be called on all the child StateManageable instances "
    "registered with this instance that are in the RUNNING or STARTING "
    "state. This causes the state to become STOPPING and on successful "
    "completion, enters the STOPPED state. All the child instances would "
    "also be in the STOPPED state. Note: Since the defined behavior is to "
    "stop all the child instances, there is not stopRecursive().") ]
uint32 Stop();
};

```

JSR77.9.5.3.2 J2EE_StateManageableLogicalElement

```

// =====
// J2EE_StateManageableLogicalElement
// =====

[Association, Description (
    "Association between the StateManageable element and the J2EE managed "
    "object.") ]
class J2EE_StateManageableLogicalElement : CIM_Dependency
{
    [Key, Description (
        "Antecedent represents the J2EEManagedObject."), Override("Antecedent") ]
    CIM_LogicalElement REF Antecedent;
    [Key, Description (
        "Dependent StateManageable element of the J2EEManagedObject referenced "
        "in the Antecedent."), Override("Dependent") ]
    J2EE_StateManageable REF Dependent;
};

```

JSR77.9.5.4 J2EE Statistics Model

JSR77.9.5.4.1 J2EE_Statistic

```
// =====
//  J2EE_Statistic
// =====
[Description (
    "The Statistics class and its sub-classes specify the data models which are"
    "required to be used to provide the performance data described by the specific"
    "attributes in the Stats class. The Statistics subclasses provide specific models"
    "to provide data about count, time and both bounded and unbounded ranges.") ]

class J2EE_Statistic : CIM_StatisticalInformation
{
// properties
[ read, Description (
    " The name of this statistic.") ]
string Name;
[read, Description (
    "The unit of measurement for this statistic.") ]
string Unit;
[read, Description (
    "A human readable description of the statisitc.") ]
string Description;
[read, Description (
    "The time the first measurement was taken represented as a long,"
    "whose value is the number of milliseconds since January 1, 1970,00:00:00.") ]
uint64 StartTime;
[read, Description (
    "The time the most recent measurement was taken represented as a long,"
    "whose value is the number of milliseconds since January 1, 1970,00:00:00.") ]
uint64 LastSampleTime;
};
```

JSR77.9.5.4.2 J2EE_CountStatistic

```
// =====
//  J2EE_CountStatistic
// =====

class J2EE_CountStatistic : J2EE_Statistic
{
[Description (
    "The CountStatistic class specifies standard count measureents.") ]
    // properties
    [ read, Description (
        " The count since the last reset" ]
        uint64 count;
    };
}
```

JSR77.9.5.4.3 J2EE_RangeStatistic

```
// =====
//  J2EE_RangeStatistic
// =====

class J2EE_RangeStatistic : J2EE_Statistic
{
[Description (
    "Specifies standard measurements of the lowest and highest values an"
    "attribute has held as well as its curent value.") ]
    // properties
    [ read, Description (
        " The highest value this attribute has held since the beginning of the measurement.") ]
        uint64 HighWaterMark;
    [ read, Description (
        " The lowest value this attribute has held since the beginning of the measurement.") ]
        uint64 LowWaterMark;
    [ read, Description (
        "The current value of this attribute.") ]
        uint64 Current;
    };
}
```

JSR77.9.5.4.4 J2EE_BoundaryStatistic

```
// =====
//  J2EE_BoundaryStatistic
// =====

class J2EE_BoundaryStatistic : J2EE_Statistic
{
[Description (
    "The BoundaryStatistic class specifies standard measurements of the upper"
    "and lower limits of the value of an attribute.") ]

// properties
[ read, Description (
    "The upper limit of the value of this attribute." ) ]
uint64 UpperBound;
[ read, Description (
    "The lower limit of the value of this attribute." ) ]
uint64 LowerBound;
};
```

JSR77.9.5.4.5 J2EE_TimeStatistic

```
// =====
//  J2EE_TimeStatistic
// =====

class J2EE_TimeStatistic : J2EE_Statistic
{
[Description (
    "Specifies standard timing measurements".)]

// properties
[ read, Description (
    "The number of times the operation was invoked since the"
    "beginning of this measurement." ) ]
uint64 count;
[ read, Description (
    "The maximum amount of time taken to complete one invocation"
    "of this operation since the beginning of this measurement." ) ]
uint64 MaxTime;
[ read, Description (
    "The minimum amount of time taken to complete one invocation"
    "of this operation since the beginning of this measurement." ) ]
uint64 MinTime;
[ read, Description (
    "The sum total of time taken to complete every invocation since"
```



```

        “the beginning of this measurement. Dividing TotalTime by”
        “count will give you the average execution time for this operation.”) ]
uint64 TotalTime;
};

```

JSR77.9.5.4.6 J2EE_BoundedRangeStatistic

```

// =====
// J2EE_BoundedRangeStatistic      (Association)
// =====
[Association, Description (
    “ The BoundedRangeStatistic class aggregagates the attributes of RangeStatistic”
    “and BoundaryStatistic and provides standard measurements of a range that”
    “has fixed limits.”) ]
class J2EE_BoundedRangeStatistic : CIM_RelatedStatistics
{
    [Override (“Range”), Description (
        “The J2EERange Statistics for the object.”)]
        J2EERangeStatistics REF Range
    [Override (“Boundary”), Description (
        “The J2EEBoundaryy Statistics for the object.”)]
        J2EEBoundaryStatics REF Boundary;
};

```

JSR77.9.5.4.7 J2EE_ResourceCountStats

```

// =====
// J2EE_ResourceCountStats          (Association)
// =====
[Association, Description (
    “J2EE_ResourceCountStats defines the association between a J2EE_Resource”
    “and its corresponding J2EE_CountStatistics.”)]
class J2EE_ResourceCountStats: CIM_Statistics
{
    [Override (“Count”), Description (
        “The J2EE_CountStatistics for the resource.”)]
        J2EE_Resource REF Count
    [Override (“Resource”), Description (
        “The Resource object associated with this J2EE_CountStatistics object.”)]
        J2EE_CountStatistics REF Resource;
};

```

JSR77.9.5.4.8 J2EE_ResourceRangeStats

```

=====
// J2EE_ResourceRangeStats          (Association)
=====
[Association, Description (
    "J2EE_ResourceCountStats defines the association between a J2EE_Resource"
    "and its corresponding J2EE_RangeStatistics.")]
class J2EE_ResourceRangeStats: CIM_Statistics
{
    [Override ("Range"), Description (
        "The J2EE_RangeStatistics for the resource.")]
        J2EE_Resource REF Range
    [Override ("Resource"), Description (
        "The Resource object associated with this J2EE_RangeStatistics object.")]
        J2EE_RangeStatistics REF Resource;

};

```

JSR77.9.5.4.9 J2EE_ResourceBoundaryStats

```

=====
// J2EE_ResourceBoundaryStats        (Association)
=====
[Association, Description (
    "J2EE_ResourceBoundaryStats defines the association between a J2EE_Resource"
    "and its corresponding J2EE_BoundaryStatistics.")]
class J2EE_ResourceBoundaryStats: CIM_Statistics
{
    [Override ("Boundary"), Description (
        "The J2EE_BoundaryStatistics for the resource.")]
        J2EE_Resource REF Boundary
    [Override ("Resource"), Description (
        "The Resource object associated with this J2EE_BoundaryStatistics object.")]
        J2EE_BoundaryStatistics REF Resource;

};

```

JSR77.9.5.4.10 J2EE_ResourceTimeStats

```
//=====
// J2EE_ResourceTimeStats          (Association)
//=====
[Association, Description (
    "J2EE_ResourceTimeStats defines the association between a J2EE_Resource"
    "and its corresponding J2EE_TimeStatistics.")
class J2EE_ResourceTimeStats: CIM_Statistics
{
    [Override ("Time"), Description (
        "The J2EE_TimeStatistics for the resource.")
    J2EE_Resource REF Time
[Override ("Resource"), Description (
    "The Resource object associated with this J2EE_TimeStatistics object.")
    J2EE_TimeStatistics REF Resource;

};
```

JSR77.9.5.4.11 J2EE_EJBCountStats

```
//=====
// J2EE_EJBCountStats              (Association)
//=====
[Association, Description (
    "J2EE_EJBCountStats defines the association between a J2EE_EJB"
    "and its corresponding J2EE_CountStatistics.")
class J2EE_EJBCountStats: CIM_Statistics
{
    [Override ("Count"), Description (
        "The J2EE_CountStatistics for the resource.")
    J2EE_EJB REF Count
[Override ("Resource"), Description (
    "The EJB object associated with this J2EE_CountStatistics object.")
    J2EE_CountStatistics REF EJB;

};
```